

Test Concept for Requirements in Data Storage System for Automated Driving - DSSAD

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Abstract

Due to the introduction of standards for vehicle data recorders in order to regulate and support the advancement of self-driving vehicles, it becomes essential to formally specify the activities and procedures for testing the EDR and DSSAD functions in accordance with the requirements stated in the published standards adopted by the further referenced markets. This document presents an explanation of the models that establish the methodological bases to support the proposal in the context of a safety application development, and then determines the testing cases for possible scenarios concerning the target requirements of each standard. Finally, an example for a particular scenario is proposed in order to show the use of Evolutive Algorithms to formulate automatically test samples for a specific test case. The main objective is to create a pragmatic test specification based on the international standards for the Chinese, US and European markets for EDR and DSSAD.

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List of Abbreviations

ADS	Automated Driving System
AEBS	Advanced Emergency Braking System
ALKS	Automated Lane Keeping Systems
AV	Autonomous Vehicle
AVS	Autonomous Vehicle Systems
BAS	Brake Assist System
BSM	Basic Safety Message
CAL	Cybersecurity Assurance Levels
CAN	Controller Area Network
CSMS	Cybersecurity Management System
DDT	Dynamic Driving Task
DoS	Denial of Service
DSSAD	Data Storage System for Automated Driving
EA	Evolutionary Algorithm
DoIP	Diagnostic over IP
EDID	EDR Data Identifier
E/E	Electrical/Electronic
ECUs	Electronic Control Units
EDR	Event Data Recorder
ESC	Electronic Stability Control
FSR	functional safety requirement
FW	Firmware
HARA	Hazard Analysis Risk Assessment
HiL	Hardware-in-the-loop
HSR	Hardware Safety Requirement
ISO	International Organization for Standardization
ms	millisecond
MRM	Minimum Risk Maneuvre
NVMe	Non-Volatile Memory Express
OBD	On-board diagnostics
OEM	Original Equipment Manufacturer
ODD	Operational Design Domain
SAE	Society of Automotive Engineers
SOTIF	Safety of the Intended Functionality
SSR	Software Safety Requirement
SSD	Solid-State Drive
STRIDE	Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege
SUT	System Under Test
TAF	Target Attack Feasibility
TARA	Threat Analysis and Risk Assessment
TB	Terabyte
TSR	Technical Safety Requirement
UNECE	United Nations Economic Common for Europe
USB	Universal Serial Bus
V&V	Verification and Validation
VRU	Vulnerable Road User

1 Introduction

One of the biggest challenges in the implementation of autonomous driving vehicles on public roads is how to resolve the driving responsibility when in case of an accident. This problem led to the implementation of software systems as DSSAD or EDR-AV to compile with the new data requirements for this scenario. In this subject the main tool available before the emergence of autonomous driven vehicles for crash analysis was the function EDR (Event Data Recorder), which purpose in vehicles is equivalent to the black box in airplanes.

EDR-AD (Event Data Recorder for Automated Driving) or DSSAD (Data Storage System for Automated Driving), have been designed for vehicles equipped with systems qualified from the third level of autonomous driving [1] and its functionality extend EDR's to cover the related scenarios and requirements.

With the entrance of DSSADs [2], new requirements have been also introduced for EDRs that entered into force in 2022[1], and therefore the assessment tests to validate whether a data recording system complies with the legislation in a given market need to be re-evaluated or extended as appropriate.

1.1 Problem Statement and Scope

In order to ensure whether a particular DSSAD implementation is compatible with respect to the legal requirements of a given market, the test concept must, as a first step, establish the functional requirements directly from the govern regulations. For this purpose, only functional specifications concerning EDR or DSSAD given within each legislative standard of the Chinese, US and European markets are analyzed in order to compile just the requirements which tests are suitable to carry out in the Safety Electronics department at IAV Fahrzeugsicherheit. In this respect, only tests that can be performed within the context of a HiL test bench with a simulated environment for the ECU on which the target function runs[2] will be considered in this document. Testing of external systems to the ECU test domain, such as systems in the cloud testing or in-vehicle functions testing are not in the scope of this research. Therefore, possible examples that are interrelated in the standards, but are not included in this test concept are:

ADS general requirements, automated lane keeping systems (ALKS), minimal risk maneuvering (MRM), advanced emergency braking system (AEBS), electronic stability control (ESC), brake assist system (BAS) or emergency steering function (ESF).

[1] EDR would be mandatory in China for new vehicle types by 2022, as well as a centralized server would collect the data to government according to GB/T 32960.3, 2016. [1]

[2] The EDR function runs normally in the airbag ECU.

Once the functional requirements have been collected, the second task is to formulate the scenarios in which these requirements might take place and the appropriate test cases that guarantee the correct behavior for the suggested scenarios. Between the new scenarios present in the standards which represent a breakthrough with respect to preceding EDRs are:

- Annotated Images: Several guidelines for data recording common to ADS involve the use of labeled images [3] [4] [5]. Among other recommendations for image acquisition, 360-degree coverage around the vehicle [4] or guided by the movement of the vehicle and the direction of impact in an accident [5] (8.4 Note2) are desirable. In addition, some suggestions indicate the need to enlarge the image quality of the contour for a better analysis of the elements entering or exiting it [5](8.4 Note3) and the possibility of using interior images to determine any sudden physical limitation of the driver that may have been present at the time of impact [6]. This document, however, takes into consideration only those requirements that are specified within a law, otherwise the technical description of the test for a requirement that is not defined, but merely proposed within the guidance of a good implementation, would not be valid. Thereby, the Chinese regulation [3] prescribes some elements that must be tested according to an appropriate image recognition strategy in order to cover this kind of scenarios.
- Storage space: Figure 1 illustrates three types of data storage in AVS: cache memory for computing system, permanent data storage in AVS like HD-map or event data recorder (EDR), and external data storage for AVS validation and verification [1]. In total, data storage might be approximately more than 100 GB of data per hour. Whatever is expected to cover the permanent data storage and, more specifically, the data logger in relation to sensor data can be expected as follows: EDR storage space is estimated between 2.5 GB for AV level 3 to 9.6 GB for AV level 4 and 5, in turn DSSAD takes between 28 to 105 GB for AV level 3 and 105 to 400 GB for AV level 4 and 5. [1]
- Cloud based applications: To deal with the expected size of data, the implementation of its storage can rely on the use of cloud storage [1] [2], this alternative is however still under development and therefore will not be included in this test specification. Besides that, the Chinese regulation proposes a centralized E/E architecture for the data encryption protocol where access control is provided through a government website [3].
- DoIP protocol application: As the Chinese specification [3] specifies the Diagnostic over IP (DoIP) to connect with Data Extraction tool, the use of this communication

protocol must therefore be explained to describe the requirements regarding the method for DSSAD data extraction in the tests.

- Cybersecurity: this concern in the form of privacy and authentication is addressed for data retrieve in the Chinese specification [3]. Hence this topic must be reviewed to set the appropriate method for testing those requirements.

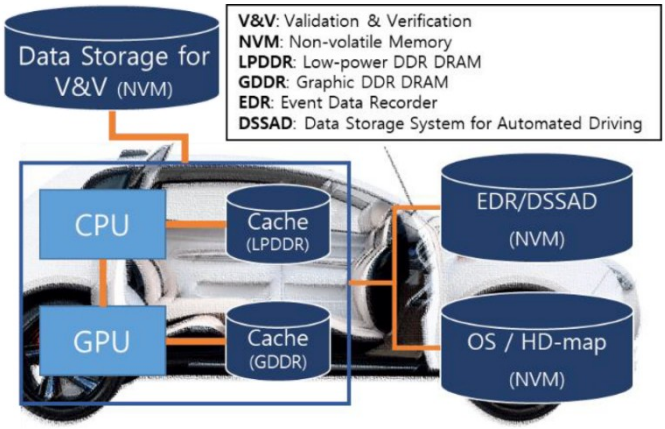


Figure 1 Data storage architecture for autonomous vehicles

1.2 Purpose

In this research paper is developed a test specification for DSSAD compliance according to the specifications of the Chinese, American and European markets. However, it just determines what is to be tested and how, the elements and product functions to be tested and the test tasks to be performed, beyond that scope, said, the test implementation and coordination as well as the aforementioned exceptions in the description of the problem are left apart. In other words, when and who is responsible for the eventual execution of the test does not fall within the scope of the present work. Figure 2 shows the goal of the present document compared to the whole scope of a test conception.

The proposal for the testing process is framed within the context of the norm ISO 26262 to justify the suggested testing methodologies and its requirements compilation is made under common requirements of the related markets in order to be able to compare the status of each legislation.

It is important here to reiterate, that this test concept is purely theoretical in the sense that the proposal only considers the requirements of the legal standards and the methods for its testing process, while the implementation in terms of software, hardware, human resources and time planning are not within the scope of this work.

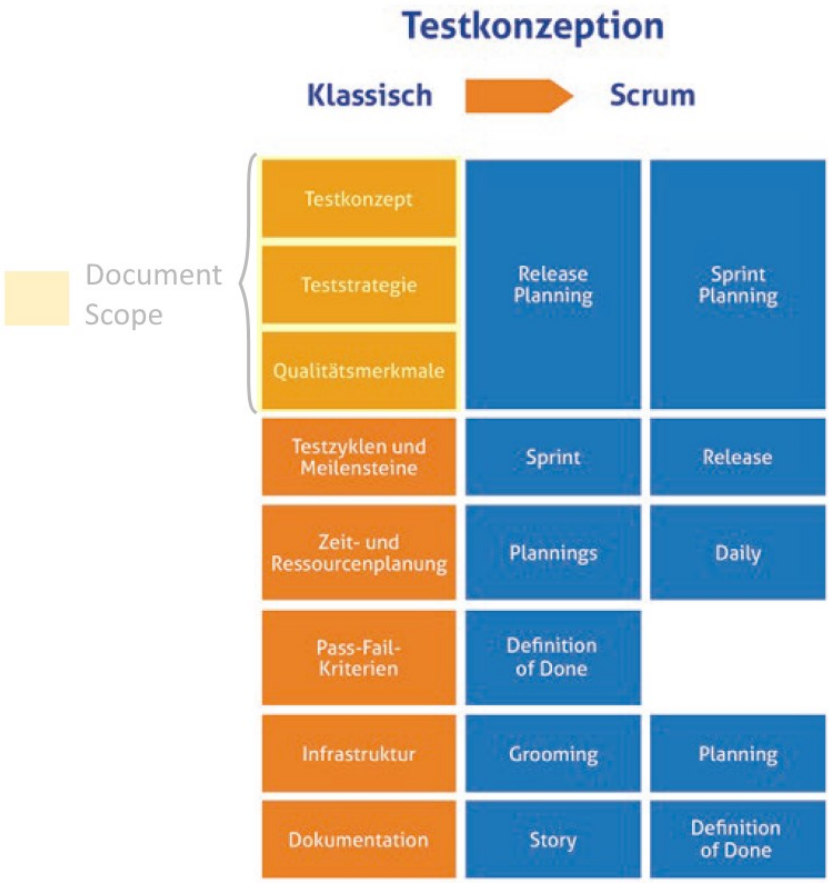


Figure 2 Scope of the document in relation with other phases of a test concept [7]

1.3 Structure of the Document

This document is divided in four chapters:

The first chapter describes as an introduction, the motivation that drives this research, the main challenges to be solved, the way the project is set up, its limitations and scope.

The second chapter goes through the basics of the test concept. First of all, it is made clear what is to be tested; what is the EDR and DSSAD function, their differences, the most important aspects for their operation and the vehicle aspects relevant for the testing proposal. On the other hand, ISO 26262 is presented as the context for the development of safety-related E/E systems, which grounds the methodologies used in the proposal and sets out this proposal within the V-Model application development.

The third chapter is itself the test concept proposal for EDR and DSSAD systems in compliance with the current legislation of Chinese, USA and European markets.

The proposal consists of the following three sub-sections:
The first step to validate the operation of the hardware under test is to describe a data input test to check that the ECU responsible for the operation of the EDR and DSSAD functions responds correctly to inputs in a HiL test environment.
The central part of the test concept focuses on the functional requirements of the standards. The test methodology is based on the ISO 26262 standard, proposing for each requirement the scenarios where it can take place and within the proposed scenarios the relevant test cases are then listed.
Finally, a concrete example is given for the use of evolutionary algorithms in the automatic generation of test samples.

In the fourth chapter the conclusions and results of the research and the subsequently test concept proposal are given.

2 Fundamentals

This chapter aims to explain the concepts involved in technical specification and the place that verification for compliance with government regulations represents within the development of vehicle applications. In turn, this engineering framework provided by a regulatory body such as the International Organization for Standardization (ISO) establishes test strategies for testing the technical and functional requirements related to safety-critical vehicle components.

2.1 ECU

Definition 2.1(ECU)

According to [8], due to the dynamic development of computing units in vehicles, the term used for ECUs in the second half of the last century varies from "electromechanical ECUs," "electrical ECUs," and "electronic (fully programmable) ECUs." This latter term is used for the physical implementation of an embedded system.
In mechatronic systems, an ECU can vary in size and performance. From single-chip devices for simpler tasks to multiprocessor systems and its sensors/actuators can also form a single unit.

2.2 Bus System

ECUs in a vehicle receive information from connected sensors and other control units. A bus system is then a network over a common transport medium to interconnect ECUs and other devices in a vehicle, this way in contrast to point to point connections presents significant improvements in terms of cost, reliability, weight and maintenance.
For the sake of this research, the following Bus systems will be considered:

2.2.1 CAN (Controller Area Network)

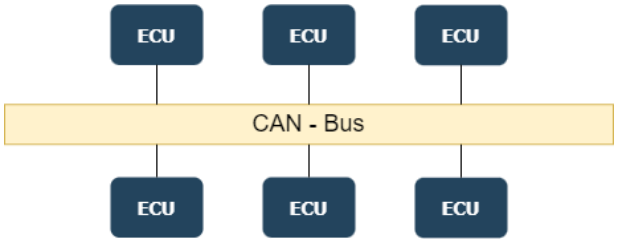


Figure 3 CAN Bus topology.

Because of its low cost and high speed, the CAN is particularly convenient in vehicle systems and is the most widely used BUS interconnection system in this type of system.

The protocol is specified in the following standards:
ISO 11898-1 CAN protocol specification.
ISO 11898-2 High-speed CAN.
ISO 11898-3 Low-speed CAN.

Data transmission speed

Although factors as transceiver latencies have to be taken into account in the transmission speed, high-speed CAN networks can achieve data transfer up to 1 Mbps with a bus length of 40m and a maximum of 30 nodes. In low-speed CAN networks, the maximum data transmission speed is 125 kbit/s with a maximum length of the transmission line of 500 m.

Topology

The Figure 3 shows a single classic line bus topology according to ISO 11898. Many CAN Bus networks can also operate as separate domains with a centralized gateway (GW) to encapsulate possible remote attacks in a sub-network.
This protocol uses a CAN transceiver and a CAN controller as the required interface within the control units for communication in the CAN network. The message transmission uses a differential signal pair, formed by the CAN high line (CANH) and the CAN low line (CANL).

	High-Speed-CAN	Low-Speed-CAN
logic 0	2 Volt	
logic 1	5 Volt	0 Volt

Table 1 Voltage for logical value in CAN

The Table 1 describes the differential voltages assigned to the logic values. The logic 0 is transmitted dominantly. In order to avoid reflections and thus interference during signal transmission, there is a terminating resistor of 120Ω at each end of the CAN network.

Communication principle

ISO 11898 defines a multi-master, message broadcast architecture for CAN bus systems. This means that each node of a CAN network operates in the same way on the network, ensuring that only one message can be transmitted over the network at any time. Within this scheme, each node can initiate a data transmission to all other nodes at any time, without following a predefined timing scheme and depending on the message identifier (ID) each receiving node discards or takes the message. In this way, the protocol does not require much configuration to add new nodes to the network, nor a protocol for the connection. The message identifier also determines the priority of the message.

Frame Types

ISO 11898-1 defines the following frame types:

- Data Frame: Data Frame (Figure 4) is used for normal data transmission. A maximum of eight bytes of user data can be transmitted.
- Remote Frame: Remote Frame is used to request data from any CAN node. It is structured exactly like a data frame without the data field.
- Error Frame: Error Frame enables the signaling of errors detected during communication. In this case, the transmission of the message in progress is interrupted. An error frame consists of only 2 fields: Error Flag and Error Delimiter.

Because vehicle communication is event-driven, the remote frame is rarely used. In particular, the data frame represents the main focus of interest for this work. The first bit of the data frame, the SOF (Start of Frame) bit, is always transmitted in a dominant form for network synchronization, while the network is in a bus idle state (recessive). The Arbitration Field contains the identifier for frame prioritization and addressing, the RTR bit indicates the type of frame (Data or Remote Frame). Next comes the IDE bit, which provides information about the format of the identifier: an identifier in standard format is 11 bits long, while the extended format consists of 29 bits. In the extended format, the RTR bit is replaced by the SRR bit, which is transmitted recessively. The two bits r1 and r0 have no meaning and are transmitted in dominant form. The length of the data field, that is, the number of user data bytes, is transmitted in the DLC (Data Length Code). This length ranges from zero to eight bytes. The user data is secured with the help of the cyclic redundancy check (CRC). The checksum required for this is transmitted as a CRC sequence followed by a recessively transmitted CRC delimiter. Based on this checksum, the receivers decide whether to acknowledge positively (dominant ACK bit) or negatively (recessive ACK bit). The ACK delimiter is always transmitted recessively. The transmission of a data frame ends with 7 recessive bits.



Figure 4 CAN Data Frame

2.2.2 DoIP

Ethernet in Automotive

Although Internet Protocol (IP) and Ethernet research in automotive is still underway in academia and major OEMs, it has several advantages with respect to other BUS standards. On one hand, Ethernet-based standards such as EthernetAVB, TTEthernet have addressed the known drawbacks of managing real-time functionality over Ethernet, and on the other hand, Ethernet's low cost, rapid deployment, security requirements, power efficiency and 100Mbps data rate offer most of the characteristics needed to cope with the ever-increasing number of new ECU nodes and data traffic in vehicle networks. Moreover, Ethernet provides TCP/IP stack to enable remote connectivity over WAN, as well as remote diagnostics, location-based applications and navigation capabilities. [9]

DoIP Standard

DoIP is an Ethernet based diagnostic protocol which was introduced to use in automotive systems. In particular, to replace regional vehicle diagnostic standards for emission control, the International Organization for Standardization (ISO) has introduced the Internet Protocol (IP) for communication between on-board and off-board diagnostic systems with the creation of the World Wide Harmonized On-Board Diagnostics (WWH-OBD) standard. Diagnostics over Internet Protocol specifies Ethernet as the physical layer and the Internet Protocol (IP) for diagnostics and reprogramming of automotive ECUs. The Table 2 shows the DoIP reference to the OSI model.

OSI 7 Layers	WWH-OBD Document Reference
Application Layer	ISO 14229-1 (UDS on IP) ISO 27145-3 (UDS on CAN)
Presentation Layer	Vehicle Manufacture Specific
Session Layer	ISO 14229-2
Transport Layer	ISO 1300-2 (DOIP)
Network Layer	ISO 1300-2 (DOIP)
Data Link Layer	ISO 1300-2 (DOIP)
Physical Layer	ISO 1300-3 (DOIP)/IEEE 802.3

Table 2 DoIP within the OSI Model reference

Diagnostics over IP (DoIP) has been standardized in ISO 13400 in following parts:

- ISO 13400 Part 1: General information and used case definition.
  Description of use case and the scenarios for the communication between testing equipment and vehicles based on an Ethernet connection (LAN and WLAN).
- ISO 13400 Part 2: Network and Transport Layer services.

Specification of the requirements for diagnostic communication between external test equipment and vehicle electronic components using Internet Protocol (IP) as well as the transmission control protocol (TCP) and user datagram protocol (UDP).

- ISO 13400 Part 3: Wired vehicle interface based on IEEE 802.3.
  DoIP support online remote diagnostics if the DoIP node has internet connectivity. The in vehicle ECU's can be accessed and diagnosed by a remote tester. IEEE 802.3 based wired or wireless connection interface standard is used.

Topology

Figure 5 shows a typical Ethernet topology with a central Gateway (GW) and Domain Controllers (DCs). The automotive Ethernet network is used as a communication backbone for the whole vehicle, so that the central GW is connected to the individual domains through DCs, which in turn control and regulate data communication between each domain and the vehicle backbone. DoIP uses a client-server architecture where the diagnostic tool acts as a client and the ECUs act as servers. As Ethernet works over a star topology, a Gateway is needed to work as intermediary, forwarding requests and responses to the respective ECUs and testers.

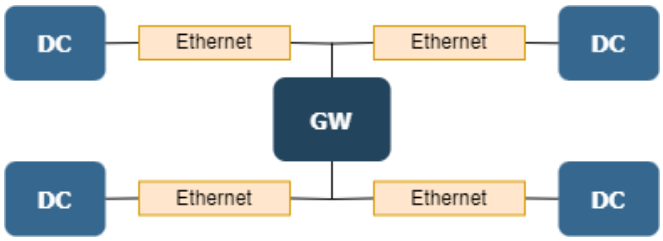


Figure 5 Ethernet Topology

Frame types

Figure 6 shows how ISO 13400 Part 3 specifies the embedded DoIP frame within the Ethernet frame. ISO 13400 Part 2 defines the generic header for DoIP based messages in 8 bytes of data containing the DoIP protocol version number, the payload type identifier and the payload length field. The DoIP payload contains in the 4 byte header the 16 bit source and destination addresses of the message, followed by the variable length payload data.

There are two types of frames used in DoIP communication:

- Diagnostic Message (UDS): This frame type is used for diagnostic purposes and is based on the Unified Diagnostic Services (UDS) protocol. The UDS protocol enables diagnostic tools to send diagnostic requests to ECUs and receive diagnostic responses from them.

- Vehicle Message (ISO-TP): This frame type is used for non-diagnostic purposes and is based on the ISO-TP (ISO 15765-2) protocol. The ISO-TP protocol enables the transmission of data between ECUs in a vehicle.

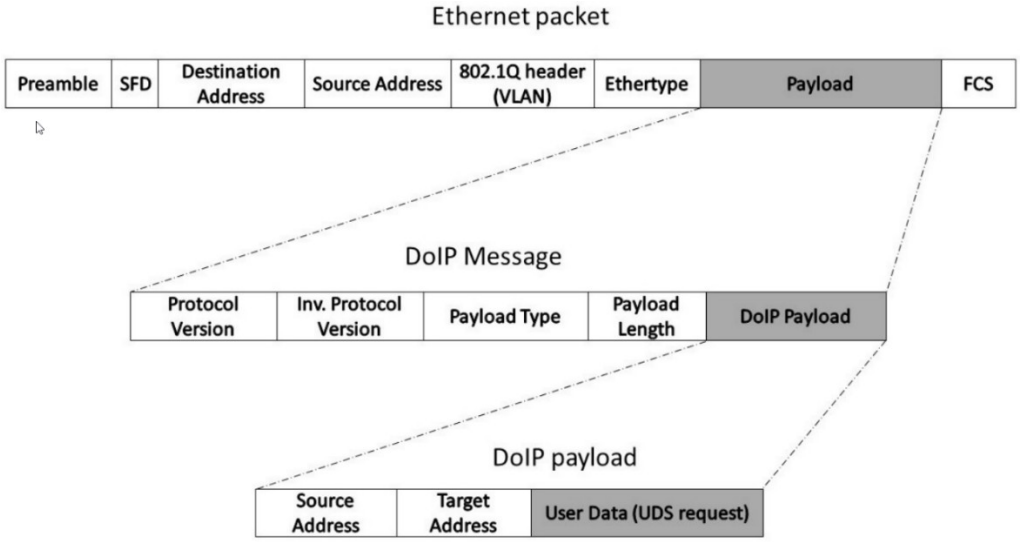


Figure 6 DoIP Frame [10]

2.3 EDR

“Event data recorder (EDR) means a device or function in a vehicle that records the vehicle's dynamic, time-series data during the time period just prior to a crash event (e.g., vehicle speed vs. time) or during a crash event (e.g., delta-V vs. time), intended for retrieval after the crash event.” [11] The purpose of an EDR is to record data related to the operation of the vehicle during a crash or other significant event. Data recorded by the EDR may include information such as vehicle speed, throttle position, brake status or whether the driver was wearing a seatbelt at the time of an event. This information can be used to determine the cause of an accident and improve vehicle safety.

2.3.1 Events

Definition 2.2 (Event)

According with AVSC [4] "event is defined as a collision or collision-like situation". Thus, a collision can be detected by several means including:

- triggers activated by meeting a speed change threshold or the deployment of any non-reversible deployable protective device.
- vulnerable road users (VRU) collision-like sensors
- In the case of AVs, actions taken in relation with by the ADS-equipped vehicle, such as pulling over, ceasing ADS operation, activation/deactivation, etc.

2.3.2 Phases

The EDR has three relevant criteria to consider when determining the accident history record:

- Start time: Or t_0. At this point the EDR freezes the continuously buffered data, the so-called pre-crash data.
- Record time: It follows when an event shall be stored in the nonvolatile memory. This decision varies subtly between regulations but, if a non-reversible restraint is deployed or the pertinent trigger is activated, then additional information taken during and after the collision is stored. If neither non-reversible restraint is deployed or the start condition is no longer fulfilled, the event shall be discarded. At this point it is also decided whether an already stored event shall be overwritten.
- End time: It determines the recording interval of the post-collision data.

The Figure 7 shows an event for which the recording action is triggered by a deployment trigger. In it the triggers are activated when the accumulated delta-V threshold is reached for each one of them. The red line represents the so-called "EDR activity state", which corresponds to the main criteria in the recording of an EDR event entry. In turn, the green line indicates the states in the timeline.

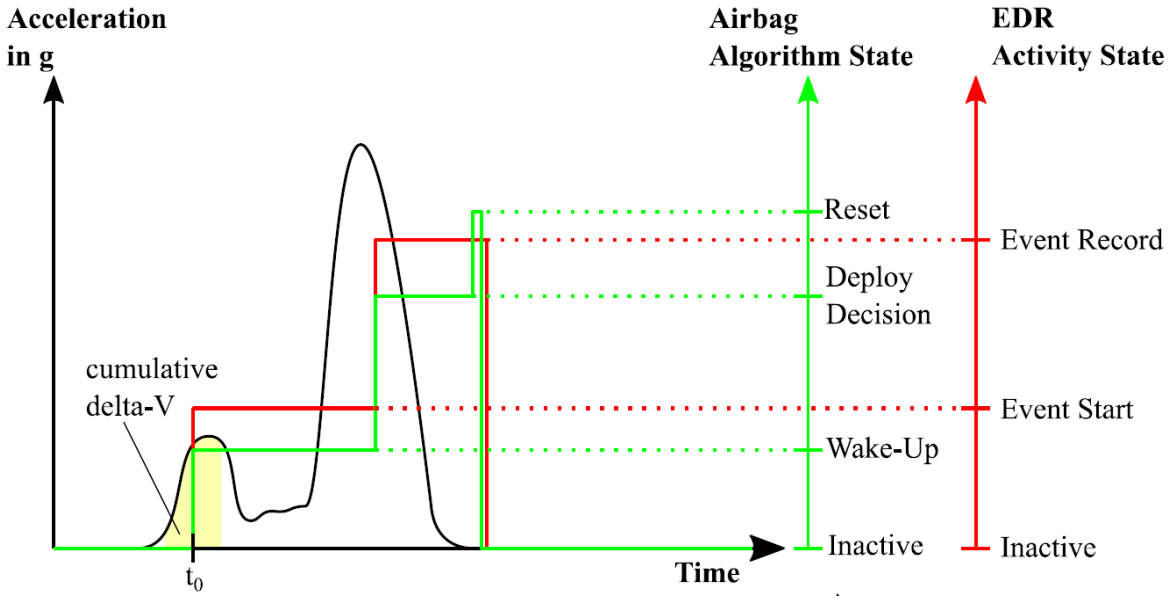


Figure 7 EDR Phases in a deployment triggered event [12]

2.3.3 Data Retrieval

The regulations specify, In case of an event, which data must be recorded and how it shall be stored. Thus, the EDR stores information before, during and after an accident in relation to the vehicle motion, i.e. accelerations and angular velocities, and the occupant status, i.e. seat

belt switch status and the deactivation status of the passenger airbag. Additionally, the standards define the recording properties (frequency, accuracy, etc.), its format (values, unit, data length, data position, etc.) and the management of this information, i.e. how to discard or overwrite events, how many of them and for how long they will be available.
In the case of EDR, the Airbag ECU is commonly the responsible to store data elements encoded in bits and bytes and lets its retrieval through a specialized data retrieval tool. The data retrieval tool interprets the encoded data and provides it in a readable format that can eventually be used for the analysis of an accident. The Table 3 quotes the kind of data managed in each Event phase.

Phase	Data provided
Driving	Vehicle acceleration Vehicle angular rate Occupant status information Vehicle status
Event (Pre-crash/In-Crash/Post-crash)	Coded Event-Data
Data Retrieval	Human Readable Data
Data Analysis	Crash Reconstruction Data

Table 3 Data provided per Phases.

2.4 DSSAD

“"Data Storage System for Automated Driving (DSSAD)" enables the determination of interactions between the ALKS and the human driver.” [13]. DSSAD is a data storage system designed specifically for automated driving systems to record data generated by sensors and cameras in an autonomous vehicle. Its purpose is not limited to accident analysis but is also used to train machine learning models and improve the performance and safety of the autonomous driving systems.

2.4.1 levels of autonomy

Most of regulations follow the categorization given by SAE J3016 [14] to define the degree of automation of the vehicles. This classification is important for the purposes of this document because it is used to define the type of data storage system on the vehicle, so the DSSAD is defined from level 3 of autonomy. Currently, ADS are defined up to level 2, so the implementation of DSSAD is still under development.

Level 0 - No Driving Automation: Where the dynamic driving task (DDT) is totally a driver’s responsibility, in other terms, there is not driving automation at all.

Level 1 - Driver Assistance: Refers to driver assistance systems installed in the vehicle that support the human driver in specific situations but does not replace him in the DDT, so that the driver must remain permanently active throughout the operation of the vehicle. Specifically, the driving function supports only either by steering or braking/acceleration, but not both simultaneously.

Level 2 - Partial Driving Automation: Refers to driver assistance systems installed in the vehicle that support the human driver in specific situations but does not replace him in the DDT, so that the driver must remain permanently active throughout the operation of the vehicle. Specifically, the driving function can support the DDT by steering or braking/acceleration simultaneously.

Level 3 – Conditional Driving Automation: is an automated driving system (ADS) that can fully perform DDT under some specific operational design domain (ODD), however, it can request the human driver to take over the DDT at any time. Example: Stop-and-go freeway traffic driving ADS feature in low-speed.

Level 4 – High Driving Automation: is an automated driving system (ADS) that can fully perform DDT under conditions without the need for a takeover-ready human driver. the ADS will automatically perform DDT fallback and achieve a minimal risk condition driving as needed. Examples: Closed campus shuttle feature, High-speed freeway cruising feature, a Valet parking, curb-to-door or a driverless operation dispatcher capable of following a pre-defined route within a confined geographical area.

Level 5 - Full Driving Automation: is an Automated Driving System (ADS) capable to perform the DDT in all circumstances manageable by a driver (no design-based weather, time-of-day, or geographical restrictions on where and when the ADS can operate the vehicle) without the need to involve a human driver in any circumstance of DDT.

2.5 EDR – DSSAD System

While both EDR and DSSAD record data related to the operation of a vehicle, they differ in their purpose and the types of data they record. EDR is primarily used to record data during a crash or other significant event, while DSSAD is used to record data generated by sensors and cameras in an autonomous vehicle to improve its performance and safety. Figure 8 summarises the main differences between EDR and DSSAD in a comparison carried out by the IWG on the Data Storage System for Automated Driving [15].

	EDR	DSSAD
Definition	… means a system with the sole purpose of recording and storing crash-related parameters and information shortly before, during and immediately after a crash [or in safety-critical events*].	…means a system with the sole purpose of recording and storing all significant interactions between the human driver and the automated driving function as well as its operation mode.
Purpose / objective	To conduct road safety analysis and assess the effectiveness of specific measures taken	To allow the attribution of liability, legal responsibility and conduct research
Scope	Step1: Vehicle categories according to R.E.3: M_1, N_1 Step 2: Vehicle categories according to R.E.3: M_2, M_3, N_2, N_3	Step1: Vehicle categories according to R.E.3: M_1, N_1 with automation level 3 or 4 and ALKS Step 2: Vehicle categories according to R.E.3: M_2, M_3, N_2, N_3 with automation level 3 or 4 and ALKS
Trigger/ events	Crashes (incl. soft objects collisions**) [and/or safety-critical events*]	Any change of conducting the driving task and/or triggered by EDR

Figure 8 Identification of differences between EDR and DSSAD [15]

2.5.1 Active and Passive safety Systems

Another way to look at the differences between EDR and DSSAD is in terms of the systems to which they are related. Although DSSAD can be presented as an evolution of EDR [16], the main source of EDR data is obtained in the context of passive safety systems like airbags, seat belts and non-reversible restraint devices, which are intended to reduce the severity of injuries in the event of a collision, while DSSAD data is conceived in the context of active safety systems like collision avoidance systems, lane departure warning systems, adaptive cruise control, and automatic emergency braking systems, which function aims to provide real-time assistance to the vehicle's operation. The Figure 9 shows the transition between both active and passive safety systems related to the course time of an accident. Before the collision active safety systems intervene in the driving process in order to prevent an accident, but once the accident has happened passive safety systems take measures to reduce its consequences.

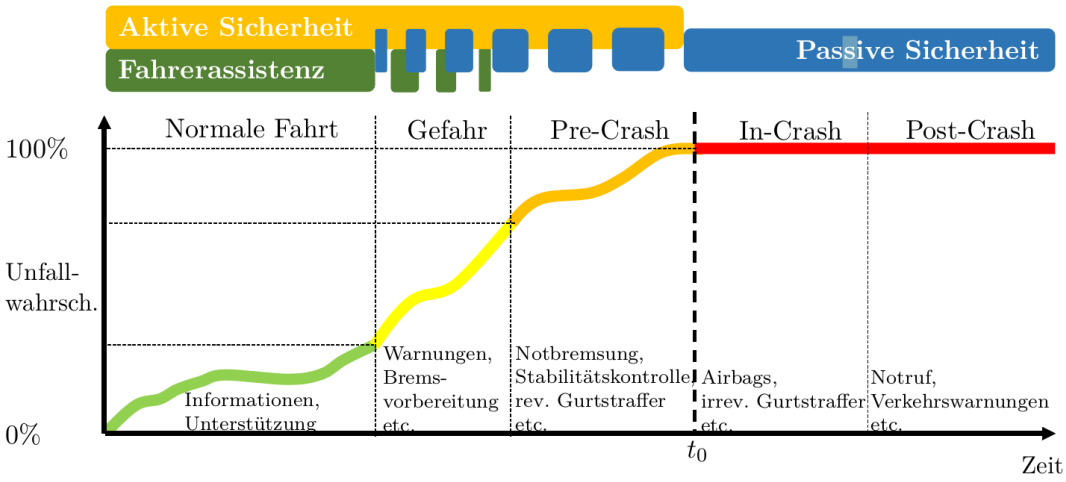


Figure 9 Course of an accident and associated safety systems [17]

Definition 2.3 (Active Safety Systems)

According to [18] Active safety refers to systems that:

- Take accident-preventing measures.
- Actively intervene in the driving process.
- Contribute to driving safety, operating safety, perceptual safety and condition safety.

Definition 2.4 (Passive Safety Systems)

According to [18] Passive safety refers to systems that:

- Take measures to reduce the consequences of an accident.
- Intervene only after an accident has been detected.
- Contribute to self-protection and counterparty protection.

In this way both technologies play a crucial role in improving the safety of autonomous vehicles by providing valuable data for analysis, evaluation, and improvement of safety systems. But DSSAD data play a more relevant role in the context to passive safety as its data can be used to understand the performance of passive safety systems, evaluate their effectiveness, and make improvements to enhance the safety of autonomous vehicles. DSSAD function is in constant operation during the use of the vehicle, while EDR function is performed only in the case of an accident.

2.6 ISO 26262

Based on the safety standard IEC 61508, the ISO 26262 covers the entire lifecycle of a safety-related electrical/electronic system installed in series production passenger cars with a maximum gross vehicle mass up to 3500 kg [19], which is the target given in the regulations

considered for this work [13] [5] [3]. Although the standard is not intended to be a guideline for the development of safety-related E/E systems and therefore does not provide specifications for its implementation, it includes a detailed description of the lifecycle of validation and verification techniques in the automotive industry, setting out the requirements for the activities and methods to be considered in each safety activity.
This standard provides a description of verification strategies and frames the present work in the context of the development of safety-related automotive electrical/electronic systems. As critical safety services, the formulation of testing strategies and the specification of requirements-based tests for EDR and DSSAD is then based on the guidelines of ISO 26262.

2.6.1 Overview

The ISO 26262 standard is organized in 10 parts:
• Part 1: Vocabulary
• Part 2: Management of functional safety
• Part 3: Concept phase
• Part 4: Product development: system level
• Part 5: Product development: hardware level
• Part 6: Product development: software level
• Part 7: Production and operation
• Part 8: Supporting processes
• Part 9: ASIL-oriented and safety-oriented analyses
• Part 10: Guidelines on ISO 26262
As a result, ISO 26262 addresses the entire life cycle of safety-relevant systems including production and operation.

2.6.2 The V-Model

Figure 10 shows the entire coverage of the standard and the part that is relevant to this work. It is noticeable, that the standard adopts a standard V-shaped model for the presentation of the development process. Thus, the process is divided into a left-hand design and development branch and a right-hand verification and validation branch. It worths to make the difference between the two terms:

Definition 2.5 (Validation)

Is it been developed the right product? Validation is concerned with the mission of why the product has been ordered or designed. It relates the first steps for the design of the product.

Definition 2.6(Verification)

Is it been developed the product right? Verification concerns the conformity of a finished product with what it was intended to be.

The Figure 10 also shows the purpose of this document in the development of applications covered by ISO 26262. In particular, the aim of the verification of the software requirements specified in the technical specifications of government regulations for EDR and DSSAD functions is not to describe tests for integrated items, but the testing for "Product Development: Software Level". This is derived from [20], as its purpose is:
"(1) to test the software units against their specifications, (2) to test particular integration levels of software against the software architectural design, and the interfaces between the software units, and (3) to verify that the software safety requirements are satisfied by the embedded software in the target environment."
In particular, (1) and (2) are the objectives driving this research.

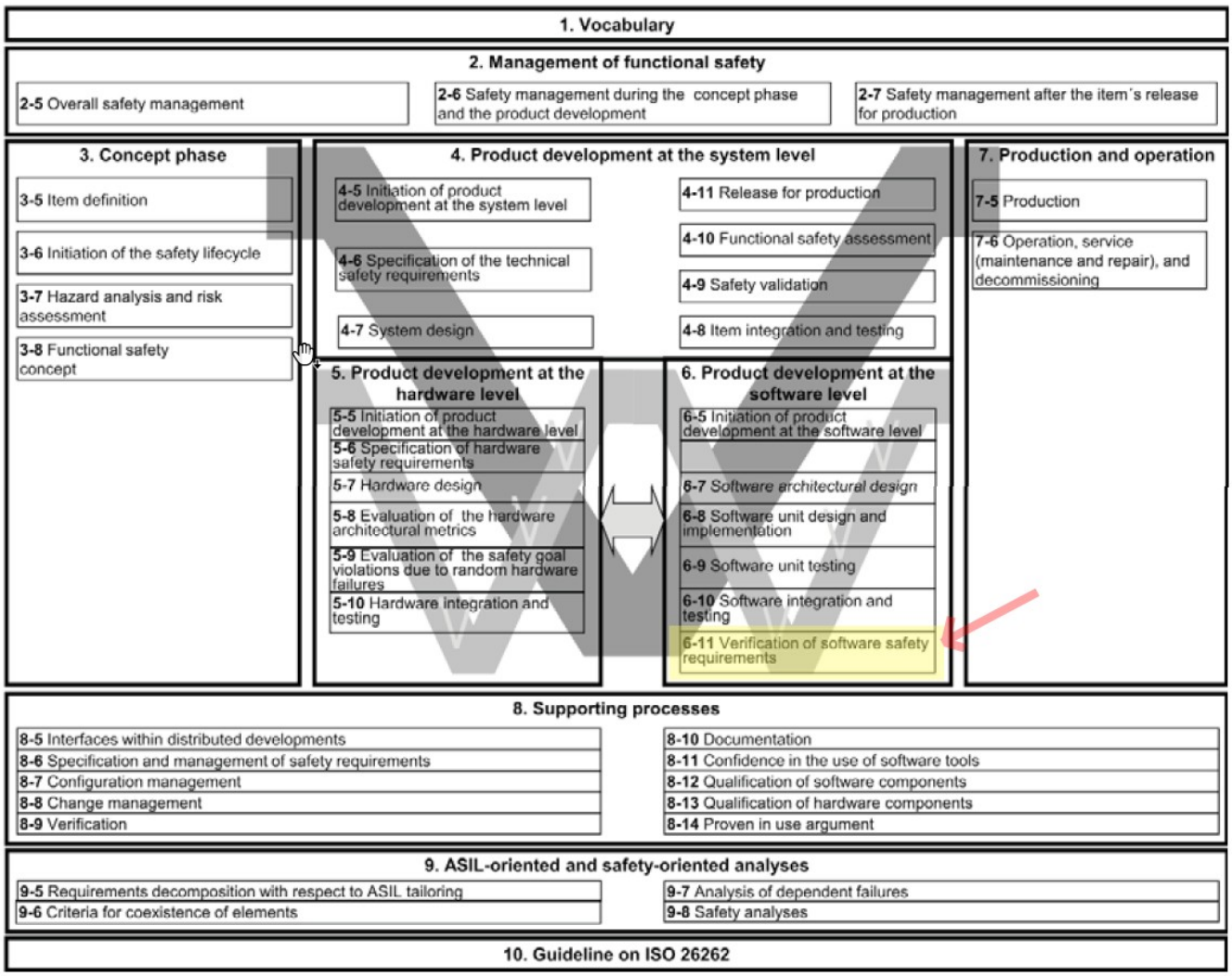


Figure 10 Structure of ISO 26262 [19]

2.6.3 ASIL Assessment

The approach to product safety compliance is based on the concept of functional safety.

Definition 2.7 (Functional Safety)

"Absence of unreasonable risk due to hazards caused by malfunctioning behavior of Electrical/Electronic systems." [19]

Based on this definition, the purpose of the safety requirements in the development of an E/E system is to guarantee the absence of unacceptable risks in the event of any malfunctioning of the system.

The process to formulate these requirements follows a hierarchical structure (Figure 11) of several phases in which a Hazard Analysis and Risk Assessment (HARA) process will be

executed according to Part 3 of the standard (Concept Phase) to specify the system safety goals that will generate the top-level safety requirements. Within the following phases, functional safety requirements (FSR), technical safety requirements (TSR), hardware safety requirements (HSR) and software safety requirements (SSR) will be successively specified and finally will detail the realization of the Technical Safety Requirements (TSR). The result of this analysis is the qualification of each development element within the ASIL (Automotive Safety Integrity Levels) frame of reference. All subsequent development phases must comply with the ASILs provided by this process.

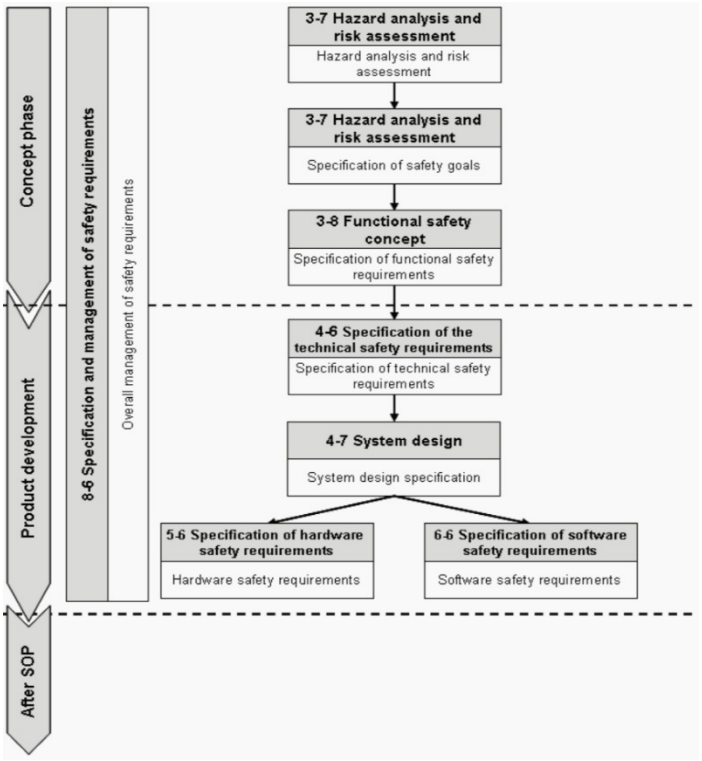


Figure 11 Structure and hierarchy of safety requirements according to ISO 26262. [19]

It is important to note that the overall risk assessment process is normally carried out by risk specialists and in the case of the EDR and DSSAD functions it involves an extensive analysis that is not contemplated within the scope of this project.

That being said, a brief qualification of EDR and DSSAD within the ASIL frame of reference is given.

Definition 2.8 (ASIL)

ASIL - Automotive Safety Integrity Level, is a four-level measure of safety requirements for a vehicle E/E system or individual vehicle components from A to D, where D is the highest.

The ASIL rate can be described as the risk derived from three factors: severity, exposure, and controllability. As a function:

Equation 1 ASIL Risk as function of severity, exposure, and controllability

Where according to [19]:

- S: Severity, potential damage for people or a system that can result of a system failure.

	S0	S1	S2	S3
Description	no injuries	light and moderate injuries	severe injuries, possibly life-threatening, survival probable	life-threatening injuries (survival uncertain) or fatal injuries

Table 4 Classification of harm or damage severity

- E: Exposure, the probability of the conditions under a particular failure can derive in a hazardous event.

	E1	E2	E3	E4
description	very low probability	low probability	medium probability	high probability
definition of duration/probability of exposure	not specified	< 1% of average operating time	1%-10% of average operating time	>10% of average operating time

Table 5 Classification of exposure probability

- C: Controllability, measure of the probability that harm can be avoided by timely reactions of the persons or systems involved.

	C0	C1	C2	C3
description	controllable in general	simply controllable	normally controllable	difficult to control or uncontrollable
definition	controllable in general	99% or more of all drivers or other traffic participants are usually able to avoid a specific harm	90% or more of all drivers or other traffic participants are usually able to avoid a specific harm	less than 90% of all drivers or other traffic participants are usually able or barely able, to avoid a specific harm

Table 6 Classification of Controllability

Although the EDR and DSSAD functions are contemplated within the vehicle safety functions, their operation does not imply any risk for the vehicle occupants, since the data processed by these functions do not intervene in the vehicle driving process and, therefore, the Severity and Exposure rating should be minimum.

2.6.4 Test methods according to ISO 26262

The purpose of this work is to verify certain functionalities of EDR and DSSAD with respect to the new guidelines dictated by the recent changes in international regulations. It is worth knowing the test methods suggested by ISO 26262 for this specific purpose. Figure 12 shows this recommendation according to which to test the correct functionality of a standard it is recommended in principle to use the "Requirements-based test" method.

Testziele auf Systemebene	Methoden	ASIL			
		A	B	C	D
Korrekte Implementierung von funktionalen und technischen Sicherheitsanforderungen	Requirements-based test	++	++	++	++
	Back-to-back test	o	+	+	++
	Fault injection test	+	+	++	++
Korrekte funktionale Performance, Genauigkeit und Timing von Sicherheitsmechanismen	Back-to-back test	o	+	+	++
	Performance test	o	+	+	++
Konsistente und korrekte Implementierung von externen und internen Interfaces	Tests of external interfaces	+	++	++	++
	Tests of internal interfaces	+	++	++	++
	Interface consistency check	o	+	++	++
	Test of interaction/ communication	++	++	++	++
Effektivität der Diagnoseüberdeckung eines Sicherheitsmechanismusses	Fault injection test	+	+	++	++
	Error guessing test	+	+	++	++
	Test derived from field experience	o	+	++	++
Robustheitslevel	Resource usage test	o	+	++	++
	Stress test	o	+	++	++
	Tests for interference and robustness and under certain evironmental conditions	++	++	++	++

Legende: o no recommendation for or against, + recommended, ++ highly recommended

Figure 12 Test methodology according to ISO 26262

Among other HIL testing methods proposed by ISO 26262, and taking advantage of the versatility of real-time simulation to test single and networked ECUs, these are the Test methods recommended for the “Verification Of Software Safety Requirements” [20]:

- Requirements-based Test

The majority of tests performed on a HIL system are requirements-based, where test specifications are first derived from the ECU requirements, which in this case are represented by the EDR functionality requirements. This results in informal but highly structured test specification documents, which include items such as identification of the specific unit under test, e.g. concrete test cases, test data and expected results for test evaluation. The next step is test execution, where results are collected and evaluated against the expected values and behaviour defined by the requirements. To provide continuity and traceability in the development and testing process, test results are linked back to the functional requirements

and commented. Test specifications are then transformed into test implementations in a test automation tool (e.g. AutomationDesk). After test execution, the results are transferred back to a requirements management tool (e.g. DOORS®).

- Fault Injection Test

A fault injection test according to ISO 26262 use of special means to introduce faults into the item. This can be done within the item via a special test interface or via specially prepared elements or communication devices. In HIL simulation, it is common to test a single ECU or a number of ECUs in exceptional situations, such as faulty operation of sensors or actuators, and errors in bus communication. Simulation of these faults can be done within the model or by using additional hardware to insert electrical faults. Fault insertion hardware is typically required to test diagnostic functionality and the response of an ECU or even the entire network to electrical faults.

Failure simulation may also be required on sensor and bus protocol pins, for example to verify/analyze the functionality of the ECU in the event of a broken CAN high wire. For sensors connected to the ECU via a differential input, disconnecting one of the input lines can stimulate floating ground effects.

- Communication And Interaction Test

A communication and interaction test according to ISO 26262 is a standard use case for HIL testing. It involves testing the communication between system elements and between the system under test and other vehicle systems during runtime against functional and non-functional requirements. These tests are commonly referred to as integration tests, as components and ECUs from different suppliers are integrated into subsystems. Vehicle manufacturers perform communication and integration tests because they are responsible for ensuring that the division of functions and the resulting distribution and use of information are correct. Suppliers perform integration/network tests for verification and also perform subsystem level tests to test or check systems during the development process.
ECU validation focuses on specification compliance and functional aspects. Simulated environments are used to simulate the environment in which the ECU under test communicates and interacts with other ECUs. Simple but appropriate models of such ECUs run in parallel with the real-time plant model, providing bus messages that are received by the unit under test. The correctness of the coding, value ranges, timing, payload and integrated safety mechanisms (e.g. CRC) is also tested. For manual or automated verification, the

vehicle bus(es) are mapped in the HIL software environment and can be continuously monitored, recorded, etc.

- Resource Usage Test

ISO 26262 describes a resource usage test that is typically performed in dynamic environments such as lab vehicles or prototypes. Testing includes power consumption and bus load, which is frequently done by using HIL simulation.

  - Power Consumption: Testing power consumption of networked ECUs is part of testing network management in the context of HIL testing. To test the positive case, the HIL simulators must behave neutrally on the bus network, using bus transceivers identical to the transceivers in the ECUs under test. To test the negative case, the sleep mode or standard operating mode of the ECUs is enforced, allowing the system behaviour to be tested for failure modes.
  - Bus Load: The bus load of vehicle networks is usually in the range of 30-70%. However, the effects caused by a heavy bus load need to be checked. Using HIL simulation, heavy bus loads are generated for testing purposes by adding a base network load (e.g. 20%). Problems caused by the additional bus load may be due to delay of message transmission or corrupted messages, which can be tested by using failure gateways that enable delaying and corrupting messages.
  - Back-to-back test: The test compares the responses of the unit under test with the responses of a simulation model to the same stimuli. In the context of HIL testing, back-to-back tests use a simulated model of the ECU or a single function, run in parallel with the HIL simulator, and then compare the results.
  - Interface Test: In accordance with ISO 26262, this methodology includes testing of analogue and digital inputs and outputs, boundary tests and equivalent class tests to check the interfaces, compatibility, timing and other characteristics of the system. Internal interfaces are tested through static and dynamic tests such as connector mating and bus communication.
  - Performance Test: a peformance test is used to verify the performance of safety mechanisms in a system, such as actuator speed, strength, and response times.
  - Stress Test: ISO 26262 stress tests are used to verify the correct operation of an item under high operational or environmental stress. Examples of stress testing include testing under high loads, extreme user inputs, extreme temperatures, humidity and mechanical shocks. HIL testing tests the input parameters of control algorithms under high operational loads, such as communication bus loads. To test ECUs under high bus loads, a "bubbling idiot" can be simulated, i.e. an ECU that

does not behave correctly on the bus, but constantly generates bus messages that significantly increase the load on the bus. Other test messages are corrupted to check the robustness of the bus communication.

2.7 HiL and Simulation Tool

Definition 2.9 (HiL Simulator)

Hardware-in-the-Loop (HIL) testing is a type of testing used in the development and validation of complex systems such as automotive, aerospace and industrial control systems. HIL testing involves the use of a software simulation tool to simulate the behaviour of various elements of the system environment, such as sensors, actuators and other external components that interact with the system under test (SUT). The process of HIL testing in a software simulated environment typically includes the following features

Development of the simulated environment: An HiL Simulator substitutes the real environment of one or more ECUs by simulating the environment and its interaction within SUT operates in real -time in a closed loop. [20]

Connecting the SUT: The test execution is performed only accessing the external interfaces of a device under test, here the airbag control unit. Therefore, testing the EDR function implies testing the airbag control unit as SUT. [12] The software simulated environment provides a controlled and repeatable test environment for evaluating the performance and functionality of complex systems. It allows thorough testing of the SUT's behaviour under different operating conditions, events and failures without the need for physical prototypes or real-world testing. Test scenarios are designed and executed in the simulated environment to simulate different operating conditions, events and failures that the SUT may encounter in real-world operation. Monitoring and data collection are essential to evaluate the performance of the SUT and validate its behaviour against expected results.

Analysis results are analysed to evaluate the performance, functionality and robustness of the SUT. Iterative testing is performed to refine the SUT design and validate its performance. The Software Simulated Environment provides a controlled and repeatable test environment for evaluating the performance and functionality of complex systems, reducing development costs, risks and time-to-market for complex systems.

3 Test Concept for DSSAD Requirements

This chapter summarises in a practical way a deliverable with the specification of activities to be carried out for the testing of coming DSSAD devices in the Safety Electronics department at IAV Fahrzeugsicherheit. The investigation of the requirements for data handling for autonomous drive systems was made up mainly of the following documents:

China

[3] GB XXXXX—XXXX: Intelligent and Connected Vehicle – Data Storage System for Automated Driving, ICS 43.040.10, CCS T36, Completion time: 15.10.2021 and Completion time: 29.07.2021

USA

[5] J3197: Automated Driving System Data Logger, Superseding J3197 APR2020, JUL2021, Issued 2020-04

[4] AVSC00004202009: Automated Vehicle Safety Consortium™ Best Practice, Revised 2021-07, SAE International.

USA-California

[21]13 CCR 228: CALIFORNIA CODE OF REGULATIONS, TITLE 13. Motor Vehicles, Division 1. Department of Motor Vehicles, Chapter 1. Department of Motor Vehicles, Article 3.8. Deployment of Autonomous Vehicles, Last amendment: Register 2021, No. 16 of 2021-04-15, Operative: 2021-04-15

3.1 General Characteristics

Based on the identified requirements, the fragments for which technical tests could be unambiguously described within the possible scenarios were extracted in order to define the test cases. On the other hand, features that would have been published only as suggestions or desirable characteristics, without being able to be contrasted under a technical specification, were not considered.

The structure adopted for the test cases consists of:

- Test description that states the purpose of the test and states what is being tested.
- Scenarios that characterize the conditions to be considered when executing the test.
- Test cases provide the specific steps to execute a test correctly.
- Approval requirements explain the desired outcomes of associated test cases.

The formulation of the test cases were formulated keeping the following principles in mind to ensure that the test specification is reliable and that the testing process is effective and efficient:

- boundary value technique: Boundary testing is a black-box testing technique that is used to check the errors at the boundaries or extreme ends of a given input domain. An input domain comprises all the possible inputs for the test case.
- Since it is not possible to test exhaustively, combinations of inputs and system states are formulated in terms of the probability of finding defects to achieve as much test coverage as possible. In addition, test cases cover both positive and negative test scenarios.
- Traceability: Traceability is the ability to trace each test case back to the requirement that it covers. This principle ensures that all requirements are covered by the test cases and that the system under test is fully tested.
- Independence: Test cases should be designed to be independent of each other so that the results of one test case do not affect the results of another test case, and test cases can be run in any order.
  Having one characteristic or feature per test also allows for more precise and accurate identification of problems or defects. When multiple features are tested simultaneously, it can be difficult to determine which feature is causing a particular problem. This reduces the complexity of the testing process, making it easier to isolate and fix defects.
- Usability: Test cases should be designed with usability in mind. The test cases should be easy to understand, easy to execute, and the results should be easy to interpret.

3.2 HiL Simulator Requirements

In order to fulfil the recording conditions required by market regulations, the software environment must be able to simulate:

China

For DSSAD Type I:

- A collision event meeting the trigger conditions in 4.1.2 [3]
- A collision risk event meeting the trigger conditions in 4.1.3 [3]
- ADS Activated
- ADS turned off, and operated by the driver
- ADS has its minimum risk strategy initiated

- ADS intervention request

For DSSAD Type II:

- ADS Activation/Deactivation
- ADS has its minimum risk strategy initiated
- ADS intervention request
- ADS malfunction signal

Software environment must also continuously stream signals for at least 8 hours for real time recording.

USA

1. Deployment data for deployment event (frontal, side, rear, rollover).
2. An impact event.
3. A pedestrian impact event.
4. A rollover event.
5. Any additional triggers defined by the ADS data logger system supplier.

Note: Triggers 1., 2. and 3. are not part of SAE J1698-1 and are therefore not defined for the EDR function.

USA-California

Autonomous technology sensor data for all vehicle functions that are controlled by the autonomous technology for at least 30 seconds before a collision with another vehicle, person, or other object while the vehicle is operating in autonomous mode.

3.3 Data Input Test

This test is required in following legislations: **China, USA, Europe**.

In HiL testing, the hardware being tested is connected to a simulation environment, which provides inputs to the hardware and monitors its outputs. The inputs to the hardware can be either software-generated or hardware-generated, and can include signals from sensors, actuators, or other devices that the hardware is designed to interact with.

During an input test, the simulation environment will generate a range of inputs to the hardware, and the outputs from the hardware will be monitored to ensure that they are correct. If the hardware is not responding correctly to the expected inputs, the input test will help to identify the issue so that it can be fixed before the hardware is deployed in a real-world environment.

3.3.1 Characteristics

The hardware to be tested is the corresponding ECU where the storage function (EDR or DSSAD) runs. The inputs are specified in the definition of data requirements Annex 1 for each market and the expected output is the correct storage of the supplied inputs in the non-volatile memory.

The input data for a test case can be seen as an array of samples in the recoding time. Each sample can be seen in turn as an array of bytes represented in the specified data type format. Figure 13 shows a such representation for the input data.

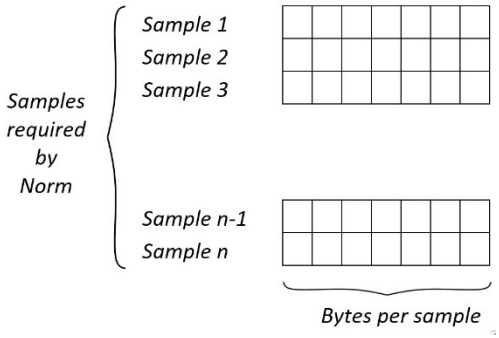


Figure 13 Data Input Representation in terms of number of samples and bytes per sample.

The number of samples follows:

$$Data\,Samples = \frac{\left| t_{Start[ms]} - t_{End[ms]} \right|}{1000} \cdot SampleRate + 1$$

Equation 2 Data Samples

Where,

- $t_{End[ms]}$: Is the time limit to sample given in milliseconds counting from (Algorithm's start).
- $t_{Start[ms]}$: Is the start time to sample given in milliseconds counting from (Algorithm's start).
- $SamplingRate_{[Hz]}$: The sample rate to sample.

*It is added one sample at the end of the equation, as one sample is taken at the beginning and one at the end of the sampling time.

3.3.2 Scenarios

The relevant differences in data input for the HiL test simulation environment are source, frequency and size of the data. It is then sufficient to test each combination of these categories.

The frequency and size of the data shall be specified in the definition of the data requirements, and the input source shall be determined depending on whether the data is generated by the simulation environment, obtained on request from the data bus or from a periodic signal transmission, generated directly by the EDR or DSSAD function, or calculated in the ECU where the memory function is performed. According to previous information, Table 7 shows the resulting scenarios according to China's standard [3].

	Data name	Origin	Storage frequency	Time interval	Unit	Data range
1	Vehicle Identification Number (VIN)	BUS	N/A	Event Trigger Time	N/A	N/A
2	Event type code	BUS/DSSAD	N/A	Event Trigger Time	N/A	0x01: AD system activated 0x02: AD system is off 0x03: Intervention request 0x04: Initiate the minimum risk strategy. 0x05: malfunction 0x06: impact 0x07: Potential impact
3	Opening of accelerator pedal	BUS	2hz	-15s~0s	%	0~100
4	Vehicle speed	BUS	10hz	-15s~0s	km/h	0~200
5	Lateral acceleration of vehicle	BUS/Sim	50hz	-15s~0s	G	-1.0~1.0
6	Vehicle roll angle rate	BUS/Sim	2hz	-15s~0s	°/s	-75~75
7	Flag of event stored completely	DSSAD	N/A	Event Trigger Time	N/A	0x00: Incomplete 0x01: Complete
8	The version number of software of the data storage system for automated driving	ID	N/A	Event Trigger Time	N/A	N/A

Table 7 Scenarios for Input test according to China's standard [3]

1. *Scenario*: Vehicle Identification Number (VIN)

The Vehicle Identification Number (VIN) is a 17-byte array represented with ASCII characters, that identifies the vehicle uniquely. Table 7 refers to this information to be received by the ECU at the time of the event trigger via the BUS system.

- Test Cases:

17 Bytes / ASCII Characters

Test Case 1	B	C	D	E	F	G	H	I	J	K	L	M	1	2	3	4	5
Test Case 2	C	D	E	F	G	H	I	J	K	L	M	1	2	3	4	5	6
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

Table 8 xVehicle Identification Number (VIN) Test Cases examples

- Test Case 1: verifies that VIN can be received and stored in the specified format.
- Test Case 2: verifies that this item is not fixed to any specific value in the hardware or in the function. This is accomplished having passed characters which are different between one sample and another.

Approval requirements: In general, "As soon as the test equipment is connected to the vehicle, Vehicle identification announcement and request is performed. This will lead the external test equipment to receive the VIN (vehicle Identification Number) of the connected

vehicle" [9]. The VIN code can be checked in the Data Extraction Process as the file name with the DSSAS event data [3] (4.6.3.3.3.D).

2. *Scenario*: Event type code

The Event type is defined by DSSAD function, whether the DSSAD function is running on the same ECU as the AD system, the AD System signals must be triggered directly in the ECU or sent via the data bus.

- Test Cases

Every case should be tested.

Test Case	Signal	Meaning	Observations
1	0x01	AD system activated	
2	0x02	AD system is off	
3	0x03	Intervention requTest	
4	0x04	Initiate the minimum risk strategy	
5	0x05	Malfunction	
6	0x06	Impact	The acceleration trigger for a collision event is given from tl simulation environment.
7	0x07	Potential impact	The acceleration trigger for a collision event is given from tl simulation environment.
8	0x08	Invalid value	The test must present an error for this test

Approval requirements: Since each type of event generates a record in the memory, it is sufficient to be able to check the corresponding record after the supply of the signals.

3. *Scenario*: Opening of accelerator pedal

The percentage of accelerator opening is given in one byte through the bus system, the input data can be represented as an array of 31 bytes, each of which is the percentage of accelerator opening at that time.

The input data shall be supplied at the required frequency until init () of the event.

- Test Cases

Figure 14 presents the test cases taken for this test.

- Test Case 1: It verifies that the required data can be properly stored.
- Test Case 2 and 3: It checks if each memory slot can vary and take new values for a random sample.
- Test Case 4: It tests possible acceptable values at the limits of the range according with the "boundary value technique". For this test it should either present an error in memory or validate the entry to the nearest valid value if a sample is outside the allowed range of values.

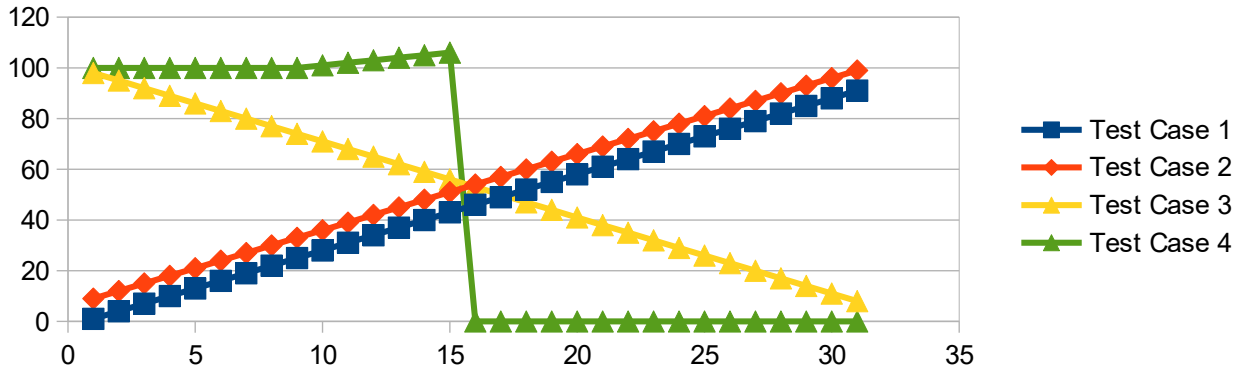


Figure 14 Test cases for "Opening of accelerator pedal" Input Test

Approval requirements: It is required to trigger an event of convenience at the time of the last sample input, to generate a record in the memory with all samples provided for the test validation.

4. *Scenario*: Vehicle speed

Vehicle speed is given in one byte through the Bus system, then the input data can be represented as an array of 151 bytes, each of which is the Vehicle speed at that time.
The input data shall be supplied at the required frequency until init () of the event.

- Test Cases

Given the number of samples for each test case input in this scenario and considering its similarity to the previous one (opening the accelerator), it is proposed to have the same number of test cases and their respective purposes for this scenario.
Approval requirements: It is required to trigger an event of convenience at the time of the last sample input, to generate a record in the memory with all samples provided for the test validation.
Data range, Storage precision and Storage accuracy must also be considered to validate this scenario in comparison with the "opening the accelerator".

5. *Scenario*: Lateral acceleration of vehicle

Since the feeding of vehicle's lateral acceleration through bus systems is only possible to a very limited extent, as ECUs check the plausibility of signals, i.e. jumps that are not physically logical are detected and recorded as errors. On the other hand, sensor inputs that are directly connected to the ECU where the data storage function runs should be simulated by a HiL simulator to take advantage of this test procedure.

Thus, in order to set this value, a software simulated environment must be used to generate these inputs in the Bus System within the course prior to the occurrence of an event.

Sampling strategy:

Another thing to bear in mind is that in a real collision, the acceleration of the vehicle varies in terms of milliseconds and, therefore, its manual prescription for 15 seconds can become an onerous task. Instead, it is suggested to set this input coarsely according to the course of some real data. Figure 15 shows the mentioned approach. The green line follows the course of a typical acceleration versus time graph (or acceleration trace or crash pulse) with a stepped curve. Another benefit of the suggested approach is that since ΔV is defined as the change in velocity between two times, if the acceleration is given as a continuous function, ΔV is then equal to the area under the acceleration curve between these times (Equation 3). Considering the suggested procedure, the calculation of ΔV for each time step would be given by the area given by the time step and the suggested value for this time interval.

$$\Delta V = \int a\,dt$$

Equation 3 Estimation of ΔV on the basis of the acceleration value

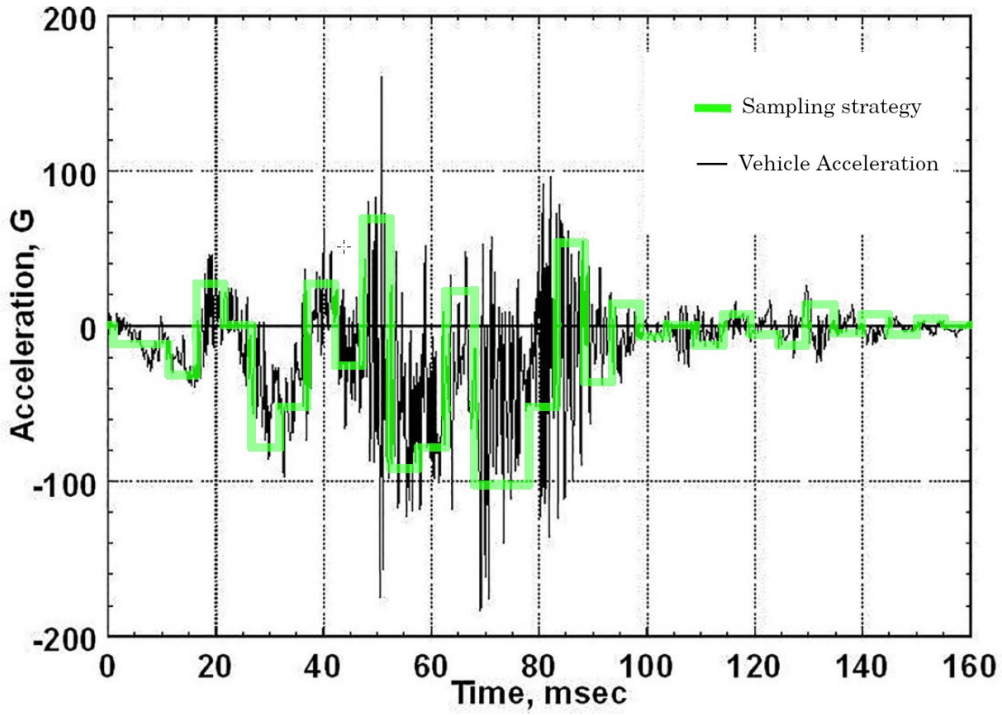


Figure 15 Raw vehicle acceleration data during a collision event [22]

Therefore, it is proposed to use the same sampling strategy used for EDR in DSSAD, specifically, a step sampling of 76 steps of 100ms each to cover the required 751 samples of the input data.

- Test Cases

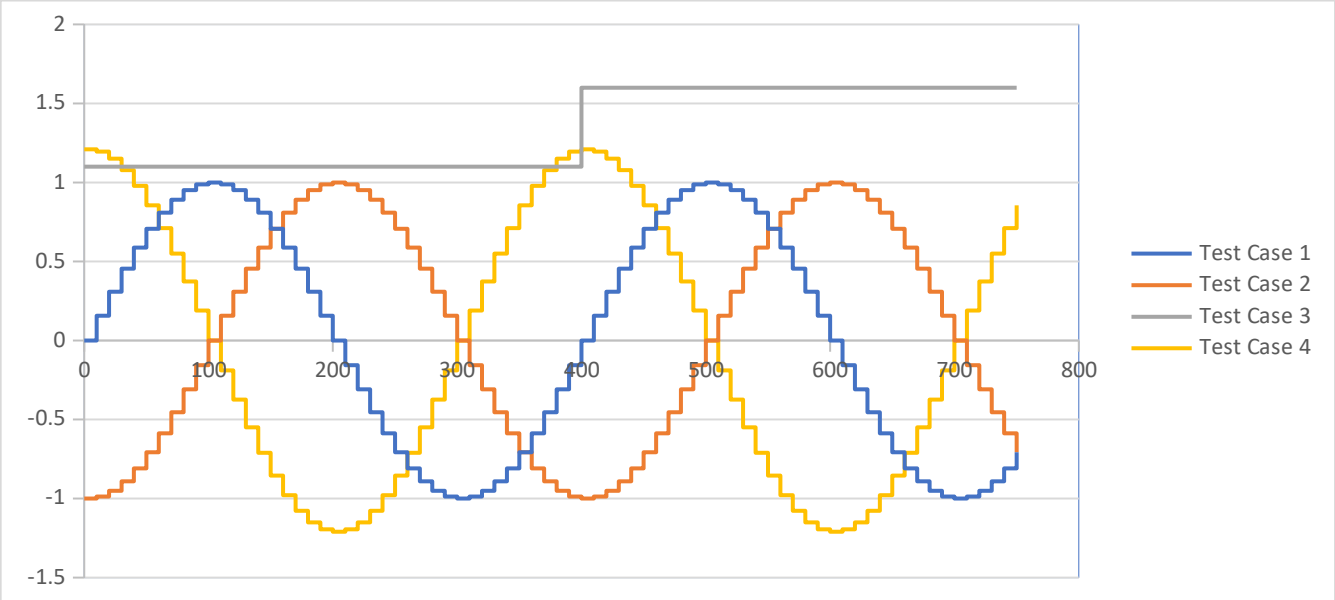


Figure 16 Lateral acceleration of vehicle Input Test Cases

The Test Cases follow the pattern outlined in the previous tests. Figure 16 shows its proposed values in relation with the sample number.

- Test Case 1: A sine curve is used to test that random data can be correctly recorded.
- Test Case 2: Using a sinusoidal curve shifted from Test Case 1 checks if each memory slot can vary and take new values.
- Test Case 3: As ECUs check the plausibility of signals, having a permanent acceleration is not physically logical and therefore this kind of signal shall be detected and recorded as an error.
- Test Case 4: Based on the "boundary value principle", some of test values given in the sine curve are outside the value range in order to generate an error in memory or validate the entry to the nearest valid value if a sample is outside the allowed range of values.

Approval requirements: It is required to trigger an event of convenience at the time of the last sample input, to generate a record in the memory with all samples provided for the test validation.

6. *Scenario*: Vehicle roll angle rate.

It is the rate at which the vehicle is changing its degree of roll. It is important to recall that this measure is a rate. As in the previous test, the roll angle rate of the vehicle must be consistent with several variables, such as vehicle speed, opening of accelerator pedal, the road in transit and the driving maneuvers. Therefore, in order to set this value, a software simulated environment must be used to generate these inputs in the Bus System within the course prior to the occurrence of an event.

- Test Cases

The 31 samples per test case are given following the same test pattern for test cases.

- Test Case 1: Random data is used to test that data can be correctly recorded.
- Test Case 2: Another set of completely different to the previous one checks if each memory slot can vary and take new values.
- Test Case 3: Based on the "boundary value principle", the test values are outside the upper value range. For this case it should either present an error in memory or validate the entry to the nearest valid value if a sample is outside the allowed range of values.
- Test Case 4: Based on the "boundary value principle", the test values are outside the lower value range. For this case it should either present an error in memory or validate the entry to the nearest valid value if a sample is outside the allowed range of values.

Approval requirements: It is required to trigger an event of convenience at the time of the last sample input, to generate a record in the memory with all samples provided for the test validation.

7. *Scenario*: Flag of event stored completely.

At some point during an event, a failure may interrupt the recording of the event. In this case, the DSSAD must be able to detect the failure and set this flag accordingly.

- Test Cases

The test cases shall cover as many possibilities as possible, where the event recording can be interrupted, here some suggestions.

- Test Case: The recording of the event is stopped due to a permanent failure in the power supply.
  In this case, the ECU hosting the DSSAD function must be disconnected from the power supply during an event. Nevertheless, data input should continue.
- Test Case: The recording of the event is briefly interrupted due to a failure in the power supply.
  In this case, the ECU power supply must be interrupted for a period around 20ms during an event. Nevertheless, data input must not stop at any time.

Approval requirements: Whether the storage interruption is permanent or partial, the event flag must be set to "0x00: Incomplete".

8. *Scenario*: The version number of software of the data storage system for automated driving

DSSAD software version number is a variable of the function itself.

- Test Cases

Depending on whether the DSSAD software can be upgraded or not, the version change can be tested before and after the upgrade.

Approval requirements: The DSSAD software version number can be verified after the storage of any event.

3.4 Tests For Types of Events

This test is required in following legislations: **China, USA**

3.4.1 Collision Event - Front/Side/Rear -

Event trigger

Test description: The purpose of the test is to determine whether or not an event is recorded in accordance with the start and finish specifications.

Scenarios

Requirement "Event shall meet the requirements of 4.1.1 in GB 39732-2020" [3].

Scenario: “For vehicles that record “delta V, longitudinal, a change in vehicle velocity in the X axis direction for not less than 8km/h within a maximum 150ms interval” [23].

- Test case: Emulation of the longitudinal delta V change (X axis direction) for less than 8km/h within a maximum 150ms interval

Approval requirements: The trigger shall not be activated, therefore it shall not track the event collision event.

- Test case: Emulation of the longitudinal delta V change (X axis direction) for 8km/h within a maximum 150ms interval
- Test case: Emulation of the longitudinal delta V change (X axis direction) for more than 8km/h within a maximum 150ms interval

Approval requirements: The trigger shall be activated, therefore the collision event shall have its t_0 at the time when the signal has been fed.

Scenario: "For vehicles that record “delta V, lateral”, a change in vehicle velocity in either the X axis or Y axis direction for not less than 8km/h within a maximum 150ms interval" [23].
Approval requirements: Test cases and approval requirements are identical to those specified for the previous scenario, however, they must be applied in the lateral direction (Y axis direction) instead of the longitudinal direction (X axis direction) to meet the requirement.

Requirement Starting time: "Shall meet requirements of 4.1.3 in GB 39732-2020" [3].
Scenario: "For systems with "wake up" occupant protection control algorithms, T_0 is the time at which the occupant protection control algorithm is activated." [23]

- Test case: Emulation of activation of occupant protection control algorithm when the system is equiped with a "wake up" occupant protection control algorithm

Approval requirements: The trigger shall be activated, therefore the collision event shall have its t_0 at the time when the signal has been fed.

Scenario: "For systems with "continuously running" occupant protection control algorithms, T_0 is the time when the cumulative Delta V of not less than 0.8km/h is reached within a 20ms interval in the longitudinal direction for a frontal/rear event or the cumulative Delta V of not less than 0.8km/h is reached within a 5ms interval in the lateral direction for a lateral impact event. As shown in Table 1." [23]
Scenario: "If EDR function is not realized through airbag controller, then clause b) shall be taken as the beginning of an impact event." [23]

- Test case: Emulation of change of cumulative Delta V of less than 0.8km/h within a 20ms interval in the longitudinal direction

Approval requirements: The trigger shall not be activated, therefore it shall not track the frontal/rear event.

- Test case: Emulation of change of cumulative Delta V of in 0.8km/h within a 20ms interval in the longitudinal direction
- Test case: Emulation of change of cumulative Delta V of more than 0.8km/h within a 20ms interval in the longitudinal direction

Approval requirements: The trigger shall be activated, therefore the frontal/rear impact event shall have its t_0 at the time when the signal has been fed.

- Test case: Emulation of change of cumulative Delta V of less than 0.8km/h within a 5ms interval in the lateral direction.

Approval requirements: The trigger shall not be activated, therefore it shall not track the lateral impact event.

- Test case: Emulation of change of cumulative Delta V of in 0.8km/h within a 5ms interval in the lateral direction.
- Test case: Emulation of change of cumulative Delta V of more than 0.8km/h within a 5ms interval in the lateral direction.

Approval requirements: The trigger shall be activated, therefore the lateral impact event shall have its t_0 at the time when the signal has been fed.

Scenario: "A deployment starting time of a non-reversible restraint". [23]
Scenario: "If c) (Previous Scenario) is adapted as the beginning of impact event, impact event data before the deployment of a non-reversible restraint shall be collected and recorded as well". [23]

- Test case: Emulation of deployment of a non-reversible restraint

Approval requirements: The trigger shall be activated, therefore the collision event shall have its t_0 at the time when the signal has been fed.

Scenario: "The second stage (or higher stage) firing of airbag shall not be taken as the beginning of an impact event, and shall not trigger another EDR record". [23]

- Test case: Emulation of second stage (or higher stage) firing of airbag during the event signaled by the first firing stage of airbag.

Approval requirements: The second stage (or higher stage) firing of airbag shall not start a new event, therefore it shall not track another event for this reason.

- Test case: Emulation of second stage (or higher stage) firing of airbag after the end of the previous event signaled by the first firing stage of airbag.

Approval requirements: The second stage (or higher stage) firing of airbag shall not start a new event, therefore it shall not track another event for this reason.

Requirement Ending time: "Shall meet requirements of 4.1.4 in GB 39732-2020." [3]
Scenario: "For "wake up" occupant protection control algorithms, T_end occurs at the moment when the occupant protection control algorithm resets itself. If this condition is not met until the end of record period, T_end could be defined as the time for data point recorded at last" [23].

- Test case: Emulation of occupant protection control algorithm reset after a given Event starting time.

Approval requirements: Event shall reset at the time of the signal.

- Test case: no signals shall be emulated until the end of the end of record period (any method can be used for this purpose).

Approval requirements: Event shall reset at the end of record period.

Scenario: "If EDR function is not realized through airbag controller, then b) (Scenario 2) shall be taken as the end of an impact event event". [23]

- Test case: Emulation of longitudinal cumulative Delta V shift, within a 20ms interval at less than 0.8km/h.
- Test case: Emulation of lateral cumulative Delta V shift, within a 5ms interval at less than 0.8km/h.

Approval requirements: Event shall reset at the time of the signal.

- Test case (d): no signals shall be emulated until the end of record period (any method can be used for this purpose).
- Test case: Emulation of airbag deployment to set starting time of the event, but no other signals until the end of record period.

Approval requirements: Event shall reset at the the end of record period.

Storage Trigger

Test description:

This specification applies only for DSSAD Type I in 4.2.2.3 [3]. The signals for both start and end time, are fed through the signal simulation software and then it is verified whether the recording has been started or ended accordingly.

Requirement Starting time: "15s before starting point of event or the instant of activation of automated driving system, whichever is later" [3]

- Test case: Using only the signals necessary to initiate the event.

Approval requirements: The last 15s before starting point shall be recorded

Scenario: Emulating the following signals later than 15s before t_0

- Test case: Activation of automated driving system

Approval requirements: Recording time shall start at the signal time.

Requirement Ending time: "5s after starting point of event or the instant of exit of automated driving system or ending point of event, whichever is earlier." [3]

- Test case: Using only the signals necessary to initiate the event.

Approval requirements: 5s after starting point of event shall be recorded

Scenario: Emulating the following signals earlier than 5s after t_0

- Test case: Exit of automated driving system
- Test case: End of event ()

Approval requirements: The recording time shall run until the time of the signal.

3.4.2 Locking conditions

Test description:
This specification applies only for DSSAD Type I in 4.2.2.4 [3]. Locking condition must be tested together with overwriting policy. Thus, each test case of blocking conditions shall be evaluated according to each test case of 3.5.2 Overwriting.

Requirement "The lock condition for impact event shall meet the requirements of 4.1.2 in GB 39732-2020" [3]. (The same as for EDR)

- Test Cases: The blocking condition shall be checked for each of the following test cases:
  - Front Impact: Emulate following signals to trigger the locking condition:
    - Non reversible restraint deployment
    - Change in X-axis direction velocity > 25km/h within 150ms interval.
  - Rear Impact: Does not apply
  - Lateral Impact: Emulate following signals to trigger the locking condition:
    - Lateral non reversible restraint deployment
    - If vehicle is not configured with lateral non reversible restraint, the test case does not apply

3.4.3 Collision risk event

Event trigger

Test description:
This specification applies only for DSSAD Type I in 4.2.3.1 and 4.2.3.2 [3]. For the test the signal simulation software must be able to emulate deceleration requests from automated driving system at several times. Then it is verified whether the event has started or ended at the time of the corresponding signal.

Scenarios

Requirement "The longitudinal deceleration requested by automated driving system is more than 5m/s2".

- Test case: Emulation of automated driving system requests for longitudinal deceleration lower than 5m/s2

- Test case: Emulation of automated driving system requests for longitudinal deceleration of 5m/s2

Approval requirements: The trigger shall not be activated, therefore it shall not track the collision risk event.

- Test case: Emulation of automated driving system requests for longitudinal deceleration higher than 5m/s2

Approval requirements: The trigger shall be activated, therefore the collision risk event shall have its t_0 at the time when the signal has been fed.

Requirement "Starting point of event: The instant when the longitudinal deceleration requested by automated driving system is more than 5m/s2 for the first time".

- Test case: Emulation of automated driving system requests at several times for longitudinal deceleration higher than 5m/s2

Approval requirements: The starting time for the collision risk event shall be set at the first request time.

Requirement "Ending point of event: The instant when the longitudinal deceleration requested by automated driving system is not more than 5m/s2 for the first time after starting point of event".

- Test case: Emulation of automated driving system requests at several times for longitudinal deceleration and no one of them is lower than 5m/s2

Approval requirements: The collision risk event shall not be terminated by any signal.

- Test case: Emulation of automated driving system requests at several times for longitudinal deceleration and at least one of them is lower than 5m/s2
- Test case: Emulation of automated driving system requests at several times for longitudinal deceleration and at least one of them is equal to 5m/s2

Approval requirements: The starting time for the collision risk event shall be set at the first request time.

Storage Trigger

Test description:

This specification applies only for DSSAD Type I in 4.2.3.3 [3].

Scenarios

Requirement "Starting point of event storage: 15s before starting point of event or the instant of activation of automated driving system, whichever is later".

Requirement "Ending point of event storage: 5s after starting point of event or the instant of exit of automated driving system or ending point of event, whichever is earlier".

Since the requirements are exactly the same as those specified for the collision event storage trigger (front/side/rear), the test of this function has already been tested on the collision event storage trigger (front/side/rear).

3.5 Data Storage Tests

This test is required in following legislations: **China, USA**

3.5.1 Capacity

Definition 3.10 (Non-Critical Safety Event)

According to 6.1.2 a) [3] Non-Critical Safety Events are the following events:

- AD system is activated
- AD system is deactivated
- AD system initiated transition demand
- AD system initiated minimum risk strategy
- When Severe failure occurs

Test description: After having save the corresponding data in DSSAD, the information stored in the data storage system for automated driving is read.

Scenarios

Requirement "The number of impact events and potential impact events that the type I data storage system for automated driving can store shall be not less than 5 and the number of non-critical safety events shall be not less than 2,500". [3]

- Test case: According to 6.5.3.1 in [3] "Before the test, the DSSAD storage space shall have stored and recorded 2500 complete non-critical safety events, and at least 5 non-critical safety events shall be triggered according to the test method of 6.1.1."

Approval requirements: "the data storage system for automated driving shall record no less than 2,500 non-critical safety events cumulatively and completely"

Requirement "Type II data storage system for automated driving shall store continuous stored data of not less than 8 hours, and the number of non-critical safety events stored shall not be less than 2,500". [3]

- Test case: Emulation of driving operation for not less than 8 hours

Approval requirements: The driving data stored by DSSAD is accumulated entirely for not less than the last 8 operation hours.

3.5.2 Overwriting

Definition 3.11 (critical safety events)

From 4.5.3.1 b) [3] “For critical safety events, impact event shall not be overwritten by potential impact event”. It follows that safety critical events are impact events and potential impact events.

Definition 3.12 (Locked event)

A critical safety events that meet the lock condition.

Definition 3.13 (Non-locked event)

A critical safety events that does not meet the lock condition.

Definition 3.14 (Lock Condition)

Any case condition specified in 3.4.2 Locking conditions

Test description:
This specification applies only for DSSAD Type I in 4.5.3.1 [3]. The signals corresponding to the events in the test cases, are fed through the signal simulation software and then the data stored in DSSAD is read.

Scenarios for DSSAD Type I in 4.5.3.1 [3]
Requirement "When the storage space of type I data storage system for automated driving is full, the following storage overwriting requirements shall be met."

Scenario: "Data of critical safety events and that of non-critical safety events shall not be overwritten by each other"

- Test case: (storage space is full for critical safety events) Emulation of more than 5 critical safety events. Then, emulation of at least one of each kind of non-critical safety event (in total not more than 2500).

Approval requirements: 5 critical safety events shall be recorded and all of non-critical events must remain in memory

- Test case: (storage space is full for non-critical safety events) Emulation of 2500 non-critical safety events plus one of each kind of non-critical safety event. Then, emulation of less than 5 critical safety events.

Approval requirements: Only the emulated critical safety event and only 2500 non-critical safety events shall be retrieved

- Test case: (storage space is full for non-critical safety events and critical safety events) Emulation of 2500 non-critical safety events plus one of each kind of non-critical safety event. Then, emulation of 5 critical safety events.
- Test case: (storage space is full for non-critical safety events and critical safety events) Emulation of 2500 non-critical safety events. Then, emulation of more than 5 critical safety events.

Approval requirements: 2500 non-critical safety events and 5 critical safety events shall be retrieved

Scenario: "For critical safety events, impact event shall not be overwritten by potential impact event"

- Test case: Emulation of 5 non-locked impact events. Then, emulation of non-locked potential impact event.
- Test case: Emulation of 5 locked impact events. Then, emulation of locked potential impact event.

Approval requirements: The 5 impact events must remain in memory and shall not be overwritten.

Scenario: "Data of critical safety event meeting the lock condition shall not be overwritten by subsequent events"

- Test case: Emulation of one critical safety event that meet the lock condition. Then, emulation of at least 5 non-locked critical safety event.
- Test case: Emulation of one critical safety event that meet the lock condition. Then, emulation of at least 5 locked critical safety event.

Approval requirements: The first critical safety event that meet the lock condition shall not be overwritten.

Scenarios for DSSAD Type II in 4.5.3.2 [3]

Requirement "When the storage space of type II data storage system for automated driving is full, the following storage overwriting requirements shall be met"

Scenario: "The data and event data stored continuously in real time shall be overwritten as per time sequence"

- Test case: Emulation of more than 8 hours of driving data.

Approval requirements: Not less than the last 8 hours of simulation shall be kept in memory. The data stored in the memory shall be the data corresponding to the last 8 hours of data fed in.

3.5.3 Power-off storage

Requirement "During the impact event, if the impact event causes the main power supply loop of the data storage system for automated driving to fail to supply power normally, the data storage system for automated driving shall store at least XXX data before the event trigger point and meet the test requirements of XXX" 4.5.4 [3].

- Test case: “Make DSSAD work normally, send triggering signal and store, disconnect power supply to DSSAD after a period of time of storage” 6.5.5.1 [3]

Approval requirements: “Power on again, read data in DSSAD, and the stored data shall meet the requirements of [1] 4.3.4” 6.5.5.2 [3].

3.6 Data Security Tests

This test is required in following legislations: **China**

The aim of cybersecurity testing is not only to ensure the communication security of the analyzed systems, but also to address factors that could lead to weaknesses and vulnerabilities in the software of automotive systems, i.e., software bugs or risks derived from the use of open-source frameworks. However, the procedures outlined here only provide test alternatives based on the safety mechanisms and requirements described in the legislation to ensure compliance.

3.6.1 Tamper-proofing, deletion, and forgery

The main risk in the case of data storage systems is that they could be manipulated to allow an attacker to alter the reason of an accident.

Requirement “When data integrity and [authenticity] are compromised, it shall be identified by technical means and stored by logs.” [3].

Scenarios

The protocol described for data collection according to [3] does not support possible overwriting or deletion of records, so that testing using the DSSAD event reader to overwrite

or delete records and subsequently verify the data manipulation record using the log cannot be considered. It is therefore necessary to provide potential vulnerabilities that could be used to simulate tampering and forgery scenarios in order to formulate the appropriate test cases for this requirement, in other words, a penetration testing technique is required to create the scenarios for this test.
It is also important to note that as the DSSAD function and associated systems have not yet been deployed on a large scale, there is no statistical data available to develop a risk analysis of the system as a whole. Another point to consider is that risks derived from vulnerabilities beyond the DoIP protocol or entry points to the internal network, such as components that a vehicle owner may voluntarily install in their vehicle, should not be considered in a compilance test.

Definition 3.15 (Penetration Testing)

In penetration testing, a tester acts from the perspective of an attacker and tries to violate security properties of a vehicle through attacks (tests) in order to uncover possible vulnerabilities. The way it is meant in this document, perform penetration testing imply simulate cyber-attacks on the DoIP communication to exploit identified vulnerabilities and gain unauthorized access, tamper with data, or disrupt communication.

Penetration testing is usually performed as a black box test with little knowledge of the system and is therefore a highly experiential activity. Providing a methodology for this process is beyond the scope of this work, so it is suggested to use a software such as the one described in [29]. An automotive port scanner tool may also cover the following functionalities:

- Gathering ECUs, Services, Subservices and Vehicle Data
- Reverse Engineering of the Routing Table
- Automatic Test Case Generation
- Fuzzing
- Vulnerability Scanning

Cybersecurity testing also includes the verification of the communication between the connected systems, it will be described in the following section.

3.6.2 Network security

According to 4.6 [3] the data extraction protocol is done via the DoIP protocol with the following considerations (*Requirements*):

- Data Extraction Port: ISO 13400-4 2013 Annex A and and the terminal arrangement is as shown in Table 10. ETH interface of the port shall meet the requirements of ISO 13400-3:2013;
- Network Layer: The IP address is IPv4, which shall meet the technical requirements of Chapter 12 in ISO 13400-2: 2019.
- Transport Layer: The transport layer shall comply with the requirements of Chapter 11 in ISO 13400-2 2019, and the port number adopted shall be "13400".
- Application Layer: IP address allocation shall meet the requirement of supporting dynamic allocation in 9.1 of ISO 13400-2: 2019.
  The message shall meet the definition of DoIP message in 9.2~9.5 of ISO 13400-2: 2019.
  When using DoIP to read the data stored by the data storage system for automated driving, according to ISO 13400-2:2019.

Terminal Serial Number	Usage	Terminal Serial Number	Usage
1	Custom	9	Custom
2	Custom	10	Custom
3	Ethernet Rx(+)	11	Ethernet Rx(-)
4	Power supply ground	12	Ethernet Tx(+)
5	Signal "ground"	13	Ethernet Tx(-)
6	Custom	14	Custom
7	Custom	15	Custom
8	Activation line	16	Power supply "positive"

Table 9 Terminal Arrangement for Data Extraction Port

Scenarios

As in the previous section, the recommended way to test compliance with ISO 13400 is using a DoIP conformance testing software such as the one offered by Vector or any other vendor, Otherwise, each layer should be tested separately considering appropriate characteristics [10]:

- Layer 1:
  - Interoperability: Link-up time, Signal Quality, Cable diagnostics;
  - Physical Medium Attachment (PMA) sublayer: Transmitter Electrical Specifications.
- Layer 2: VLAN Testing, Address Learning, Filtering of incoming frames, Time synchronization, Quality of Service, Configuration.
- Layer 3-7:
  - Address Resolution Protocol (ARP): Packet Generation, Packet Reception;

- Internet Control Message Protocol Version 4 (ICMPv4): Error Handling, ICMP Types;
- Internet Protocol Version 4 (IPv4): IPv4 Header, IPv4 Checksum, IPv4 Time to Live, IPv4 Version Number, IPv4 Addressing, IPv4 Fragments, IPv4 Reassembly;
- Dynamic configuration of IPv4 Link Local Address: Introduction, Address Selection, Defense and Delivery, Announcing an Address, Conflict Detection and Defense, Link-Local Packets Are Not Forwarded, Healing of Network Partitions;
- User Datagram Protocol (UDP): UDP Message Format, UDP Datagram Length, UDP Padding, UDP Fields, USER Interface, Introduction, Invalid Addresses;
- Dynamic Host configuration Protocol Version 4 (DHCPv4) Client: Summary, The Client-Server Protocol, Client-server interaction – allocating a network address, Client parameters in DHCP, DHCP usage, Constructing and sending DHCP messages, Initialization and allocation of network address, Reacquisition and expiration;
- Transmission Control Protocol (TCP): Connection Establishment and Basic Exercising of the State Machine, Processing and Generating TCP Checksums, Processing Unacceptable Acknowledgements and Out of Window Sequence Numbers, Processing TCP RECEIVE Calls Received from the Application Layer, Processing TCP ABORT Calls Received from the Application Layer

3.6.3 Encryption and Authentication

Requirement “The automated driving data recording system shall obtain the Safety Authentication Protocol on Data Access through SM9 Cryptographic Algorithm based on identity identification for the purpose of conducting the identity certification and authentication of data readers.” 4.7.2 [3].

Definition 3.16 (SM9 Cryptographic Algorithm)

“SM9 is a Chinese cryptography standard, which includes a set of identity-based cryptographic schemes over pairings” [24].

Requirement Security protocol comprises 4.7.2 [3]:

- Data Extraction Tool: comply with requirements in 4.4.3 [3] Data Reading Protocol and supporting interaction with the Authentication Center.
- DSSAD: Supports SM9-related algorithms, including IBE-KEM, DEM, etc. The user private key is only stored in KGC cloud other than local device.

- Authentication Center: Provides functions over internet and cloud such as user registration, authentication and interaction with KGC.
- KGC: Key Generation Center works out the user private key calculated based on master key and user identification and is able to interact with the Authentication Center.

Requirement Certification Scheme Scenarios 4.7.2 [3]:

- Online: The data extraction tool communicates directly to Authentication Center to retrieve decrypted data from DSSAD.
- Offline: The data extraction tool obtains encrypted data form DSSAD, and get separately the decryption key from Authentication Center.

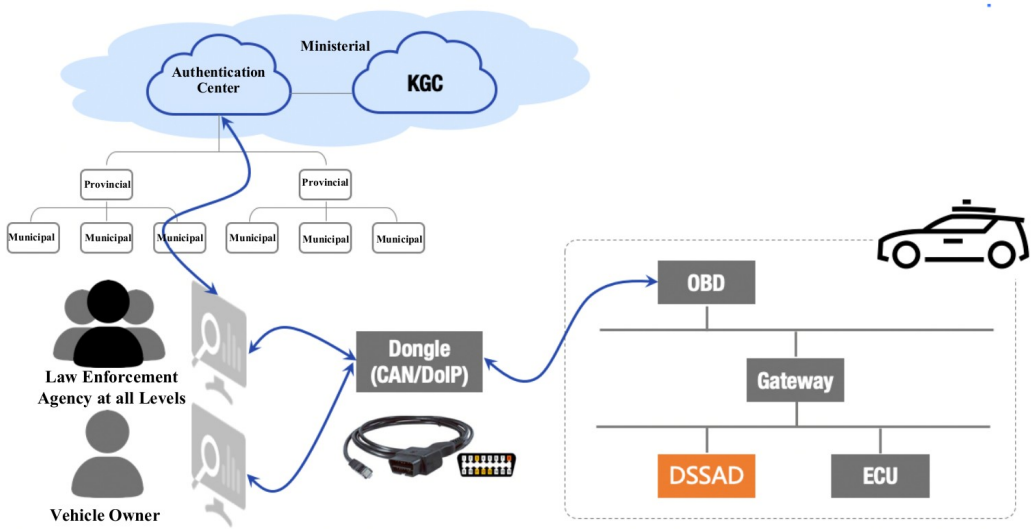


Figure 17 China Encryption and Authentication Protocol Architecture for DSSAD [3]

The Figure 17 shows how the authentication protocol is defined. There are two major components in the architecture configuration. One is in the cloud and consists of the Authentication Centre and the KGC. The other is composed of the devices used to process and store the DSSAD information and the mechanism provided for the connection in the vehicle, as well as the dedicated device for data extraction.

Scenarios

The required functionalities for each component in the encryption and authentication task are:

- KGC - Key Generation Center – (KMS)
  - User private key storage.
  - User private key estimation based on master key and user identification.
- Authentication Center

    - User registration: provides vehicle and personal information, including vehicle license, driving license and ID card, etc.
    - Identity certification: After completing the registration successfully, a string of unique characters will be generated based on certain rules.
    - Performs the authentication with KGC and obtains the private key (deB) of the user after the authentication is succeeded.
    - Calculates the Seed through the algorithm IBE-KEM-1(deB, C) based on the user private key deB obtained and the ciphertext C sent by the reading tool.
    - returns Seed to the reading tool.
- DSSAD
    - Checks if the user identification is valid.
    - Generate Seed and C (the ciphertext of the Seed) through IBE-KEM(ID user identification) and return C to the reading tool.
    - Compares the Seed saved previously with the Seed sent by the reading tool; if they are the same, the safety authentication will be passed and the user may read the data through the reading tool.
- Data extraction tool
    - Sends the user ID and requests the seed to DSSAD.
    - Accesses the Authentication Center and completes the identity certification through username and password.
    - Sends C to the Authentication Center.
    - Sends the Seed to DSSAD.

Each of which should be included as a scenario in the compliance test to ensure that they are implemented correctly and effectively.
Each scenario must be tested using each of the proposed connection schemes, providing authentication access of users with different permissions, including users such as regulatory authorities and public security departments, automobile enterprise, and vehicle owners.

The standard has not yet described how the SM9 protocol is used. Since a proposal for a conformance test would only be possible based on the definition of the implementation of each of the functionalities specified in the standard, only an assumption of how identity verification could be performed in the context of the SM9 protocol is given in this paper. In this regard Figure 18 shows a possible implementation of the authentication process via SM9 protocol.

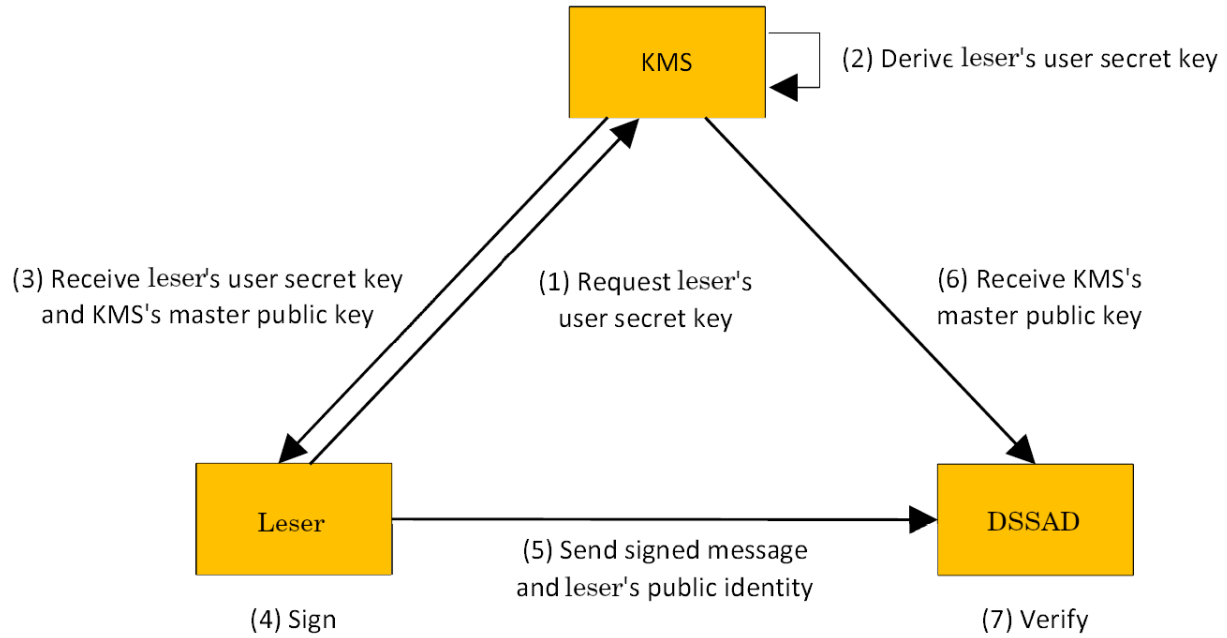


Figure 18 Implementation approach of the SM9 protocol for the authentication process [24]

- Provisioning a Data Extraction Tool involves the following steps:

1) Data Extraction Tool requests the user secret key corresponding to her public identity (e.g. the Serial Number) from the KMS.
2) The KMS derives Data Extraction Tool's user secret key from her public identity and the master secret key.
3) The KMS securely sends to Data Extraction Tool its user secret key and the master public key.

- Sending a signed message involves the following steps:

1) The sender, Data Extraction Tool, signs the message using its user private key and the master public key.
2) Data Extraction Tool sends the signed message to the recipient, DSSAD, together with its public identity.

- Verifying the signature on a received message involves the following steps:

1) DSSAD obtains the master public key from the KMS.
2) DSSAD verifies Data Extraction Tool's signature using its public identity and the master public key.

3.7 Use of evolutive algorithms to automatically generate test samples

The reason for addressing this topic is essentially to propose an evolutionary algorithm for generating test data. This method has several advantages, which will be discussed in detail below.

3.7.1 Fundamentals

The three states of the EDR function relevant to the test are:
- Inactive: Where the function records the relevant data in a ring storage capable of recording the relevant data at pre-crash time continuously.
- Event Start: It is the moment where an event occurs, this according to [11] is: "Event means a crash or other physical occurrence that causes the trigger threshold to be met or exceeded, or any non-reversible deployable restraint to be deployed, whichever occurs first." At this moment the pre-crash memory recorded in the ring storage becomes part of the event record.
- Event Record: as [11] "Record means the process of saving captured EDR data into a non-volatile device for subsequent retrieval." From this point, the recorded pre-crash and post-crash information is transferred to a non-volatile device for later retrieval.

Figure 19 shows the three states of the EDR function in relation to the Airbag deployment function.

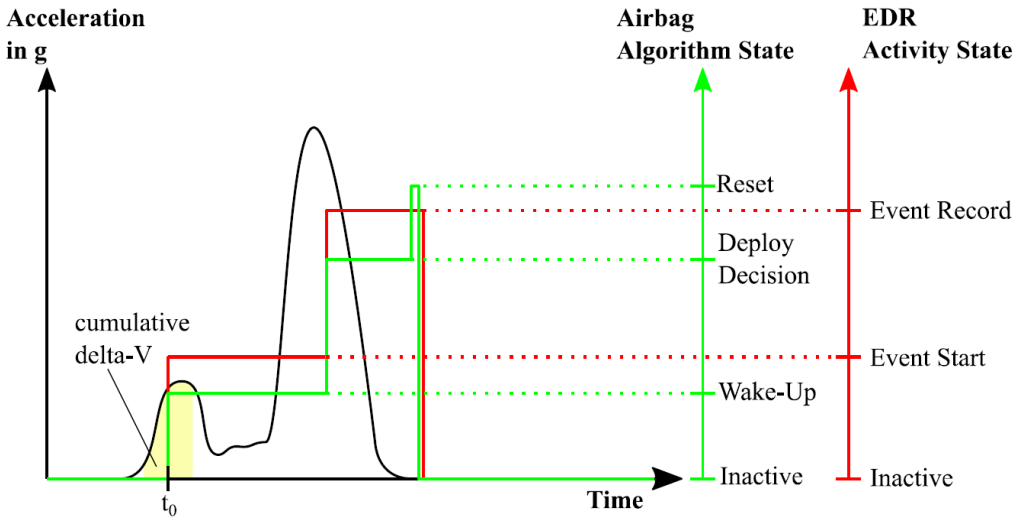


Figure 19 Exemplary mapping of crash activity to airbag algorithm states and EDR activity states. [12]

3.7.2 Problem under consideration

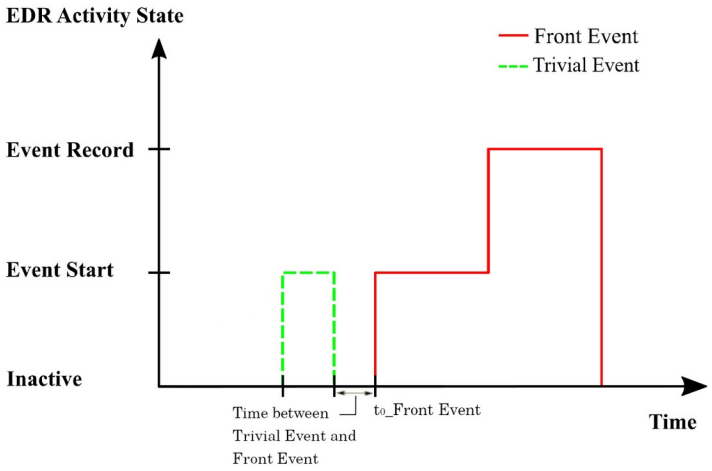


Figure 20 Diagram of a front event preceded by a trivial event.

Trivial Event: It is the case where the conditions for an "Event Start" are fulfilled and then the "Inactive" state is restored without passing through the "Event Record" state.
Front Event: It is the event where the conditions are met to state that the vehicle has had a frontal impact. In this case it goes from an "Inactive" state to "Event Start" and then to "Event Record". A possible "Event Lock" state is not examined in this analysis.

Figure 20 shows the case considered. According to [12] there is a possibility that when analyzing parallel or simultaneous events, failures may occur in which data are not recorded correctly, completely, or according to the standard. In this case, if the trivial event ends close to the beginning of the frontal event, the trivial event can lead to partial or complete discarding of the pre-crash data prior of the frontal event. The purpose of this test is to check whether all or part of the recording time prior to a front event can be lost if a previous event is discarded before or during the start of the front event.

3.7.3 Implementation

“In evolutionary testing test cases are identified and evaluated analog to the pattern of recombination and mutation of genes in biology. An initial population is changed and optimized until a certain desired characteristic is accomplished.” [25]

The testing process and the test sample generation process are parallel processes that alternate in such a way that each leads to the beginning of the other. Figure 21 shows both processes as cycles whose sequence allows to separate the phases of "test samples

creation" and "test execution with the real system", so that both mechanisms converge at the beginning and at the end of each cycle.

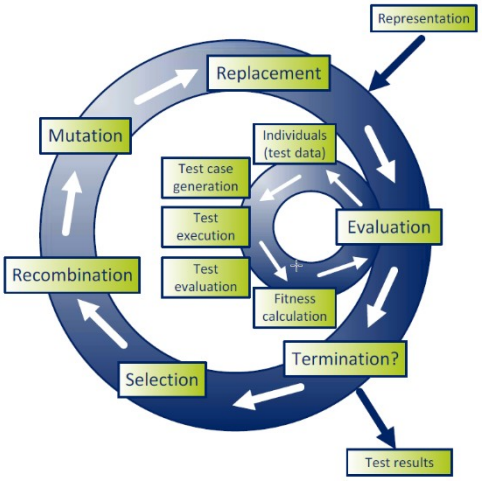


Figure 21 Use of evolutionary algorithm for the creation of test cases [25]

The inner cycle includes the phases of test data collection (test samples), test case generation, test execution and test evaluation. It represents the test execution based on the hardware-in-the-loop (HiL) environment having the input data from the corresponding airbag control unit interface.

The outer cycle shows the phases of the test sample creation process thorugh the use of an Evolutionary Algorithm. This pprocedure responds to the impossibility of generating or testing all possible combinations of events that could occur in an accident. Therefore, test data should be generated according to a policy guided by a good testing principle that fits the problem we want to address.

To systematically find the test cases [12] that most likely deliver a possible error, an evolutionary algorithm based on the following policies is specified:

- Defects cluster together: When an error is detected in a test, the probability of finding more errors of the same type increases in proportion to the proximity of the parameters used in the test setup.
- Beware of the pesticide paradox: If a significant number of tests return no error, it is very likely that no remarkable results will arise with similar parameters either.

In other words, the parameters should be changed if the tests do not give relevant results, and vice versa, if an error is detected, testing should be done near the parameters used.

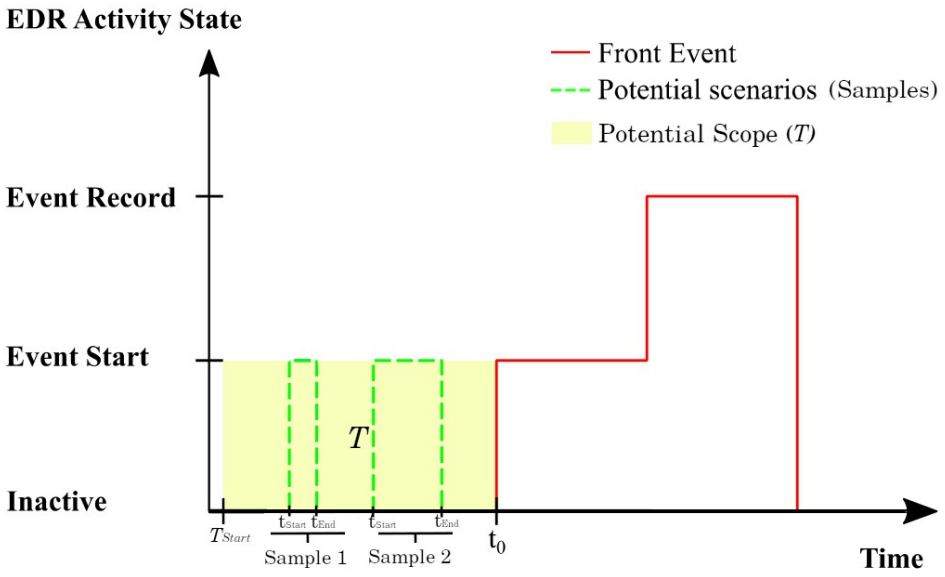


Figure 22 Parameters for EA (Evolutionary Algorithm)

The parameters used for the detection of possible failures in the pre-crash data are the start time (t_{start}) and the end time (t_{End}) of the previous trivial event in relation to the start time (t_0) of the front event.

The parameters of a test sample can be written as:

(t_{start}, t_{End}) where $t_{Start} < t_{End}$ and $t_{End} \in T$ or with specific values $t_{End} \in (T_{Start}, t_0)$

where and or with specific values

Where:

T: Pre-crash recording time established in the regulation.

T_{Start}: Instant when pre-crash recording time starts.

Figure 22 shows the parameters to optimize and its range.

The following are the key features within the formulation of the evolutionary algorithm:

- **Population-based**

A population is a group of Individuals which in an evolutionary algorithm are the possible solutions regarding the problematic condition. In turn, a single solution would be considerated as an individual, which in the current context is a possible test sample.

- **Fitness-oriented**

Each Individual has two properties Code and Quality which correspond to the “gene representation” and the “performance evaluation”. “EAs prefer fitter individuals, which is the foundation of the optimization and convergence of the algorithms.” [26]

Code (Gene representation):

Are the individual properties/genes. For the case, these are the Event parameters to be optimized by the Evolutionary Algorithm, hence, the parameters to optimize are the code or gene representation in this case: t_{Start} and t_{End} of a trivial event, that may occur prior to the frontal event. Hereafter it will be represented as the tuple (t_{Start} , t_{End}).

Quality (performance evaluation):

The fitness value is evaluated for the first time in the first iteration (generation) using the following two criteria to score each sample test

1. Following the "Defects cluster together" policy, we shall prioritize when a test sample is close to some prior test that has led to an error. Thus:

$$V_1 = \frac{\min_{i \in S} \left(a \cdot \left| t_{Start} - E_i\left[t_{Start}\right] \right| + b \cdot \left| t_{End} - E_i\left[t_{End}\right] \right| \right)}{1000}$$

Equation 4 First criteria for fitness function

with:

E: Set of test samples that have led to find an error.

S: Number of elements in *E*

$E_i\left[t_{Start}\right]$: t_{Start} of element i in *E*

$E_i\left[t_{End}\right]$: t_{End} of element i in *E*

a: Factor to weight $\left| t_{Start} - E_i\left[t_{Start}\right] \right|$

b: Factor to weight $\left| t_{End} - E_i\left[t_{End}\right] \right|$

* It is desirable to prioritize $\left| t_{End} - E_i\big|_{t_{End}} \right|$ over $\left| t_{Start} - E_i\big|_{t_{Start}} \right|$, then b must be accordingly higher than a, thus $b > a$.

* As the best fit for the sample in *E* must be taken, the minimum scored sample over E is the one to be taken for the optimization.

* If *E* is empty, then $V_1 = 0$

** The value obtained from the minimum is divided by 1000 so that the value of V_1 is comparable with the of V_2 (next criteria).*

2. The aim of the test is to explore the impact that a trivial event that ends within the pre-crash recording time established in the norm may have on a frontal event, therefore:

$$V_2 = 0 \text{ if } t_{End} \subset T$$

Otherwise:

$$V_2 = \left| t_{End} - T_{Start} \right| \text{ or } V_2 = t_{End} - t_0$$

Equation 5 Second criteria for fitness function

Thus, the Fitness Value of a test sample i is given by:

$$f_i = V_1 + c \cdot V_2$$

Equation 6 Fitness function

Where

c: Is a factor to weight V_2.

* As V_2 corresponds with the main purpose of the test, it should be correspondingly higher than a and b, therefore $c > b > a$.

- **Variation-driven**

"Individuals will undergo a number of variation operations to mimic genetic gene changes, which is fundamental to searching the solution" [26]

Selection

It is going to be used the "Roulette wheel selection" (RWS) method for selection.

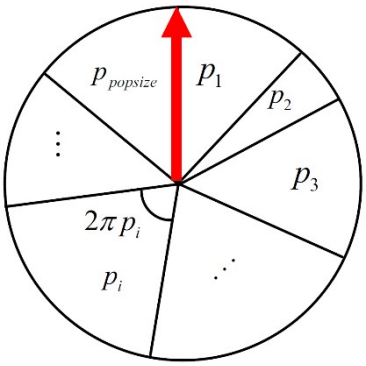


Figure 23 Roulette wheel selection - RWS

Then:

relative fitness value for individual i:

$$p_i = \frac{f_i^{-1}}{\sum_{i=1}^{popsize} f_i^{-1}}, \text{ where, } \sum_{i=1}^{popsize} p_i = 1$$

Equation 7 Relative fitness value to use RWS in selection step for EA

* The Equation 7 usues f_i^{-1} instead of f_i because f_i has the best score at its minimum zero.

And a test sample i will be selected by: $\sum_{i=1}^{k} p_i < rand < \sum_{i=1}^{k+1} p_i$, where rand is a random number between [0,1]

Mutation

In order to implement the Evolutionary Programming algorithm given in [26], the so-called "normal mutation" [26] will be implemented. Normal mutation uses the following equation to perform the mutation on a parameter i of a test sample:

$$x_i = x_i + N(0, \sigma)$$

Equation 8 Normal mutation function to mutation step in EA

3.7.4 Algorithm

The following algorithm is taken from [26] in "Solution Process of One Type of EP Implementation".

1. Initialization.
   1.1. Assign the parameters for EP, such as λ, μ, and σ.
      μ: $popsize$
      λ: The number of new created test samples from selected ones in "Selection" Step
      σ: The deviation in the normal distribution in the mutation function will alter the copy of selected samples. In order to follow "Beware of the pesticide paradox" policy, we want to have enough variance in the new test samples to explore new possibles approaches if no errors are found with the given population.
   1.2. Generate μ uniformly distributed individuals randomly to form the initial population and evaluate their fitness values. gen = 0.
2. Main loop. Repeat the following steps until gen > maxgen.
   2.1. Repeat the following operations until a new population with μ individuals has been generated. Perform a mutation (Eq. 2.12) for every gene of the individual to generate a new one.
   2.2. Calculate the fitness value for every new individual.
   2.3. Combine μ current and λ new individuals and pick the μ best ones to form a new population. gen = gen+1.

Submitting the final μ individuals as the results of EP.

The following steps illustrate its implementation, taking into account the characteristics of the case analysed in this document:

1. Random test samples creation: In order to maximize the exploration of possible errors, new random samples will be created, so that every time the test is executed, it will take new values to test. The restriction for the creation of test samples is the time range where the end of the trivial event can take place (T_{Start}, t_0).
2. Test execution and following test sample creation process.

2.1. Evaluation: After the first test execution, the test results are evaluated within the fitness function [Equation 6].
2.2. Offspring: Then, by means of Roulette wheel selection (RWS) method the samples to generate new samples (offspring) are selected.
2.3. Mutation: In this case the "Recombination" phase does not take place, so directly the new samples are mutated by means of Normal mutation function [Equation 8] for each property of the new samples (t_{Start}, t_{End}).
2.4. Mutated Samples Evaluation: The new samples are evaluated within the fitness function.
2.5. New generation: The best (*popsize*) evaluated test samples from the set of current and new test samples are taken for the next iteration of the process.

* As the goal of the optimization process is to create as many error-prone test samples as possible, there is no single goal, hence it will take a defined number of optimization cycles before delivering test samples back to the testing process.

4 Results

The results found throughout the test specification process for compliance with government requirements for Event Data Recording (EDR) and Data Storage System for Automated Driving (DSSAD) are given for each of the processes included in this thesis:

4.1 Applicable Regulations Identification

EDR

At the time of elaboration of this document, completed and under development regulations in China, UNECE (Europe), North American Region (NAR), Japan, Korea, Uruguay, Australia and New Zealand related to EDR were reviewed.

In most cases, the documents simply refer to the function of EDR within a general road safety context. Documents specific to data management and relevant elements of EDR functionality were only found in the markets of China, Europe, NAR, Japan and Korea. The most relevant for the establishment of assessment criteria being those set out in the Table 10.

China	GB39732-2020-Vehicle Event Data Recorder System-EN
UNECE	GRSG-120-03 UN Reg on EDR - 01 Series Chairs
NAR	49 CFR PART 563 EVENT DATA RECORDERS
Japan	Kokujigi 278-2008 EN 2011-12-20
Korea	MOLIT Ordinance No. 906 (2021.10.25)

Table 10 Most relevant documents per market for the specification of EDR requirements

DSSAD

In the case of DSSAD, a review was made regarding complete and under development regulations for China, UNECE (Europe) and the USA. In general, these regulations are still under development in all cases. For the US, the most relevant documents refer to best practices and recommendations for the development of these systems[3]. In the case of China, the regulation, still changes significantly from one version to another in a matter of months[4], even though it is the most specific and complete of all. For the European market and more precisely for the German market, the DSSAD requirements are found within the standards of other ADSs as for example, the standard dedicated to the standardization of Automated Lane Keeping Systems is the one that defines most of the requirements (Input Data in 3.3) considered for this work. The documents that were primarily considered for this specification are listed in Table 11.

[3] The title of AVSC00004202009 [4] is "AVSC Best Practice for Data Collection for Automated Driving System-Dedicated Vehicles (ADS-DVs) to Support Event Analysis", while SAE J3197 [5] falls as a "surface vehicle recommended practice" document.

[4] That is the case for “Intelligent and Connected Vehicle – Data Storage System for Automated Driving” [3], which version of 29.07.2021 was discarded because its differences with the one of 15.11.2021 [3].

China	[3] GB XXXXX—XXXX Intelligent and Connected Vehicle – Data Storage System for Automated Driving ICS 43.040.10 CCS T36 Completion time: 15.10.2021
Germany	AFGBV Anlage 2 Datenspeicherung[5]
USA	[5]Automated Driving System Data Logger Superseding J3197 APR2020 JUL2021 Issued 2020-04 [4] Automated Vehicle Safety Consortium™ Best Practice Revised 2021-07 SAE International AVSC00004202009

Table 11 Most relevant documents per market for the specification of DSSAD requirements

4.2 Requirements Review

The following are the most important characteristics to mention.

EDR

The regulations examined do not reveal major differences between them, however:

- The intention to extend the EDR recording period for timed data metrics from 5 seconds of pre-crash data at a frequency of 2 Hz to 20 seconds of pre-crash data at a frequency of 10 Hz within the NAR regulation [27].
- The entry into force of laws introducing mandatory application for all new vehicles from 2022 in the Chinese [1], NAR [27] and European [28] markets.
- Introduction of Vulnerable Road User (VRU) safety protection system for triggering EDR activation in Europe (5.3.3.3 of GRSG-120-03).

DSSAD

As the development of requirements is quite different between the markets, it is worth reviewing the particular characteristics of each one individually.

<u>China</u>

The review of the requirements taken for this work is mainly based on this technical specification. Notable features include the following:

- The DSSAD function includes all EDR functionalities, however, the specification of events is done for two types of DSSADs:
    - DSSAD Type I: Its functionality is basically extended from EDR as it takes the same types of events and triggers for data recording, but over an extended timeline[6]. It also uses triggers that incorporate features of an AVS, such as the ADS activation or an ADS intervention request.

[5] https://www.gesetze-im-internet.de/afgbv/anlage_2.html

[6] Start: 15s before starting point of event; End:5s after starting point of event

    - DSSAD Type II: This type of DSSAD stores no less than 8 hours of continuously recorded data during the time the automated driving system is activated. Apart from that, there are no new requirements with respect to the EDR specification.

  Both DSSAD types shall be able to store at least 2500 non-critical safety events.
- For data extraction, the DoIP protocol is specified together with the authentication protocol upon SM9 Cryptographic Algorithm, which includes a centralized cloud-based system for authentication of Data Extraction Tools.

USA

- The events defined for EDR are defined in the same way for DSSAD.
- The definition of pedestrian impact and rollover events is included based on the appropriate sensors for each case.
- The defined data elements are defined in relation to the level of automation of the vehicle.
- Event recorded data may be located in on-board systems, off-board systems (e.g., cloud storage), or both for storage and processing.
- Data readers must be in compliance with industry best practices for data integrity, such as ISO 9001:2015.
- The following characteristics are desired for visual representations:
    - 360-degree data set around the ADS-DV collected by cameras, LIDAR, radar, ultrasonic sensors, or other means.
    - ADS self-criteria can set recording frequency and enable/disable use of sensors.
    - Visual representation of salient objects and conditions leading up to an event.
    - Annotated Images.
    - The minimum pixel resolution may need to increase as the number of degrees in the field of view increases to be able to capture salient objects at the edges of the field of view.

USA-California

- Pre-crash data begins 30 seconds before a collision with another vehicle, person, or other object while the vehicle is operating in autonomous mode.

Europe

- The DSSAD is expected to store 2500 events, which is equivalent to 6 months of use.[7]

4.3 Test Cases Formulation

The resulting technical requirements are used to specify tests that certify compliance with applicable regulations. The specification is easily and effectively presented for each of the test activities and their proper and rapid implementation. For example, in cases where the standard simply requires implementation of another standard, it has referenced the latter to generate the appropriate scenarios and thus provide a complete review of the standard. In addition, the requirements have been broken down by market, taking into account the differences between them and avoiding repetition of the proposed procedures for similar requirements.

The following are the achievements developed in the development of the Test Cases:

- The test cases are based on the requirements set out in the standard.
- The test cases cover various scenarios and use cases relevant to data storage systems in autonomous vehicles.
- The test cases include the validation of:
    - Data accuracy.
    - Assurance that data is stored for the required time period.
    - Encryption and authentication mechanisms.

4.4 Evolutive algorithm to generate test samples

The analysis of the specific case for the formulation of the algorithm led to the generation of a general algorithm for its implementation, thus giving the possibility of using this method in other scenarios that typify the problems studied throughout the given formulation.

[7] Addendum 156 – UN Regulation No. 157

5 Conclusions

Although the collection of information included in this document covers legislation up to August 2022 and it is estimated that the most relevant documents on this matter have been included, it is impossible to ensure a review of all regulations concerning driving data storage. In this respect, taking into account that AV-related legislation is growing progressively, it is also expected that the legislation concerning data storage for ADS will be extended within the coming years and consequently the corresponding specification of the verification tests. It has to be recognised that even at an early stage of ADS development whose DSSAD legislation is limited to establishing whether the human or autonomous system was in control during and prior to the time of an accident, the Chinese legislation is the most advanced in relation to other markets, whose dispositions are mainly done in the form of recommendations or best practices.

Regarding the relationship between DSSAD and EDR, it has been found in the regulations examined that the EDR's functionality is contemplated within the functionality of DSSAD. However, it is presumed that the development of DSSAD will not eliminate EDR, as although there is an overlap of data stored by both functions, the difference in the volume of data created from ADS data is expected to be large enough to at least re-evaluate the way of the data not related to accidents but only to the operation of ADS is stored. Among others, is the possibility of data storage in the cloud and the provision of a new device (ECU) dedicated exclusively to process this information rather than adapting the Airbag ECU for both functionalities. On the other hand, there are recent proposals for EDR subsequent to publication of some others for DSSAD, suggesting a separate agenda for each functionality. This information is important, as the current development and test execution procedures for EDR will not be substantially affected by the emergence of DSSAD.

When formulating test cases, it should be recalled that it is impossible to consider all possible scenarios and events that might reveal a bug. As Dijkstra famously said in 1972: "Program testing can be a very effective way to show the presence of bugs, but is hopelessly inadequate for showing their absence". In this respect, not all of the methodologies given by ISO 26262 have been used in the specification of the tests, but having structured the test proposal on the specification of the ISO 26262 standard for the particular case of Verification of Software Safety Requirements in a HiL environment, facilitates the understanding of the target and purpose of the specified tests. This background also provides alternatives in the execution of the tests and gives an idea of more test possibilities than those already proposed.

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7 Anex 1

The following tables show a representation of the data required by each piece of legislation for the DSSAD discriminated by source, frequency and size of the data.

China

USA

	Data name	Origin	Storage frequency	Time interval	Unit	Data range
1	Time of day	BUS	1 sample	-5 seconds	YYYY:MM:DD:HH:MM:SS	1 second
2	ADS Action – ADS Requested Hazard Flashers	BUS	2 Hz	-5~0	N/A	Yes or No
3	ADS Action – ADS Requested Gear	BUS	4 Hz	-5~0	N/A	As applicable per transmission (typical values are Drive, Park, Reverse, Neutral, etc.)
4	ADS Action – ADS Requested Lateral Vehicle Motion Control	BUS	5 Hz	-5~0	Manufacturer Specific	Manufacturer Specific
5	Other Sensor and Input Data	BUS	Manufacturer Specific	-5~0	Manufacturer Specific	Manufacturer Specific
6	**Salient Object Attributes**	BUS				
7	Data Record Trigger Type	BUS/Sim*	1 sample	0 seconds relative~time zero	N/A	Threshold,Pedestrian Device, Rollover, ADSPerceived Collision, ADS Unable to Perform DDT

Germany

	Data name	Origin
1	Fahrzeugidentifizierungsnummer	BUS
2	Positionsdaten	BUS
3	Anzahl und Zeiten der Nutzung sowie der Aktivierung und der Deaktivierung der autonomen Fahrfunktion	BUS
4	Anzahl und Zeiten der Freigabe von alternativen Fahrmanövern	BUS
5	Systemüberwachungsdaten einschließlich Daten zum Softwarestand	BUS Möglicherweis DSSAD
6	Umwelt- und Wetterbedingungen	BUS
7	Vernetzungsparameter wie beispielsweise Übertragungslatenz und verfügbare Bandbreite	BUS Möglicherweis DSSAD
8	Name der aktivierten und deaktivierten passiven und aktiven Sicherheitssysteme Daten zum Zustand dieser Sicherheitssysteme sowie die Instanz die das Sicherheitssystem ausgelöst hat	BUS/ECU
9	Fahrzeugbeschleunigung in Längs und Querrichtung	BUS/Sim
10	Geschwindigkeit	BUS
11	Status der lichttechnischen Einrichtungen	BUS
12	Spannungsversorgung des Kraftfahrzeugs mit autonomer Fahrfunktion	BUS/DSSAD
13	von extern an das Kraftfahrzeug gesendete Befehle und Informationen	BUS